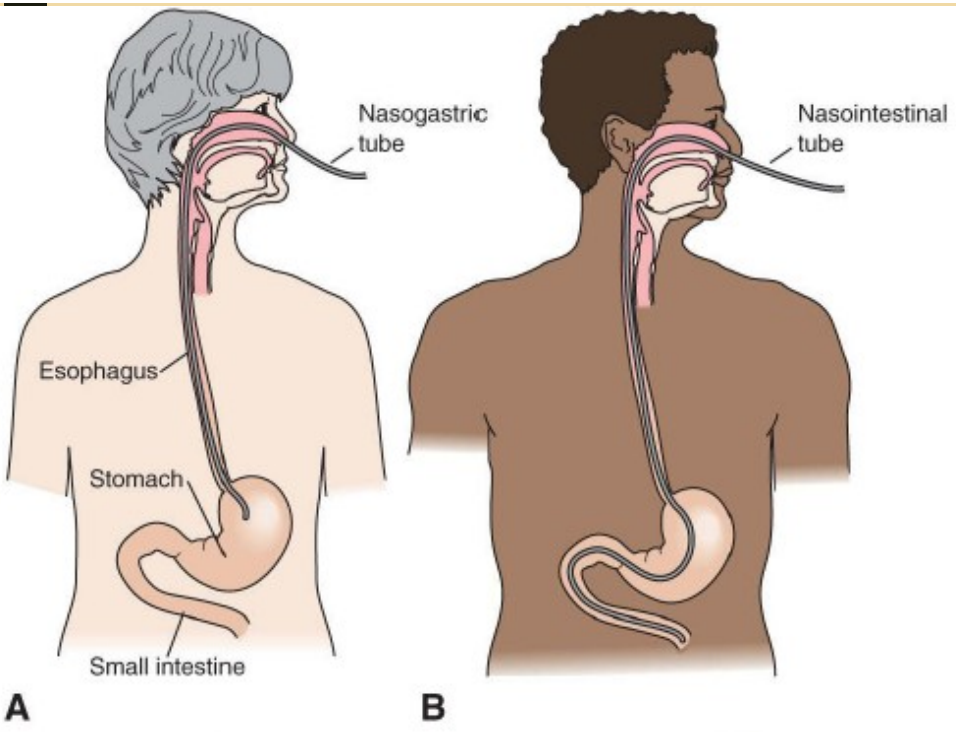
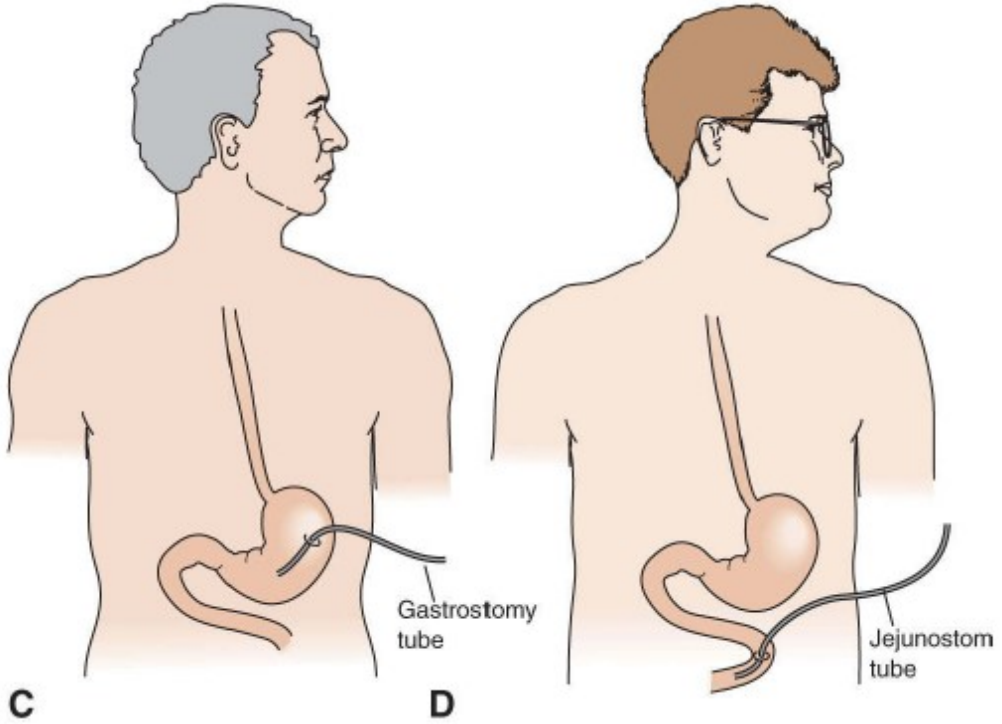
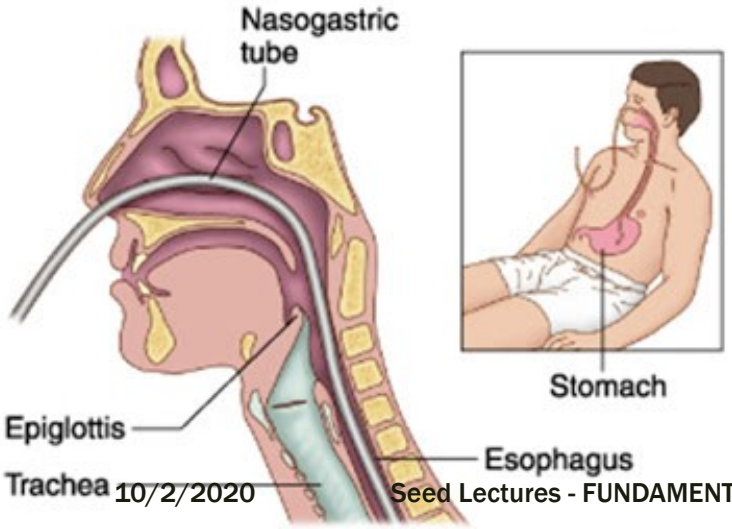
GASTROINTESTINAL INTUBATION

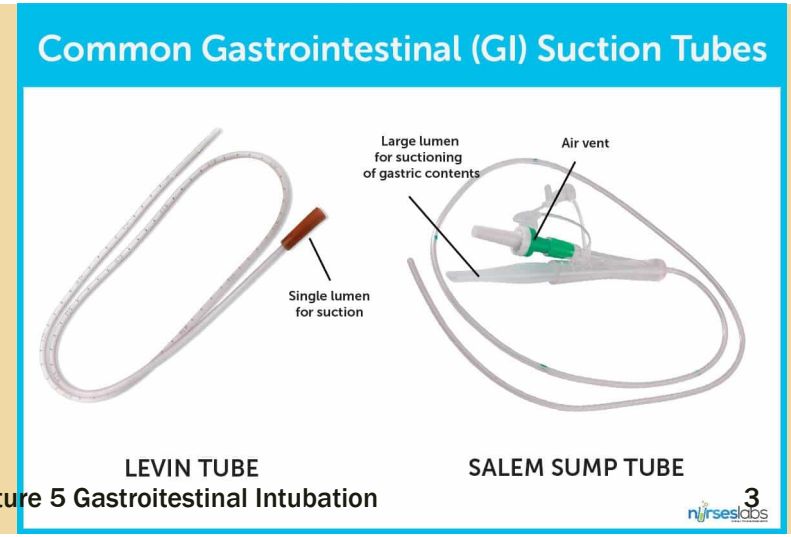
INTUBATION

- **Intubation generally** means the placement of a tube into a body structure
- **Orogastric intubation** (the insertion of a tube through the mouth into the stomach),
- **Nasogastric intubation** (the insertion of a tube through the nose into the stomach), and
- **Nasointestinal intubation** (the insertion of a tube through the nose to the intestine) are performed to remove gas or fluids or to administer liquid nourishment.
- A tube also may be inserted within an **ostomy** (a surgically created opening).


Nasogastric tube
Esophagus
Stomach
Small intestine
A
Nasointestinal tube
B


Gastrostomy tube
C
Jejunostom tube
D

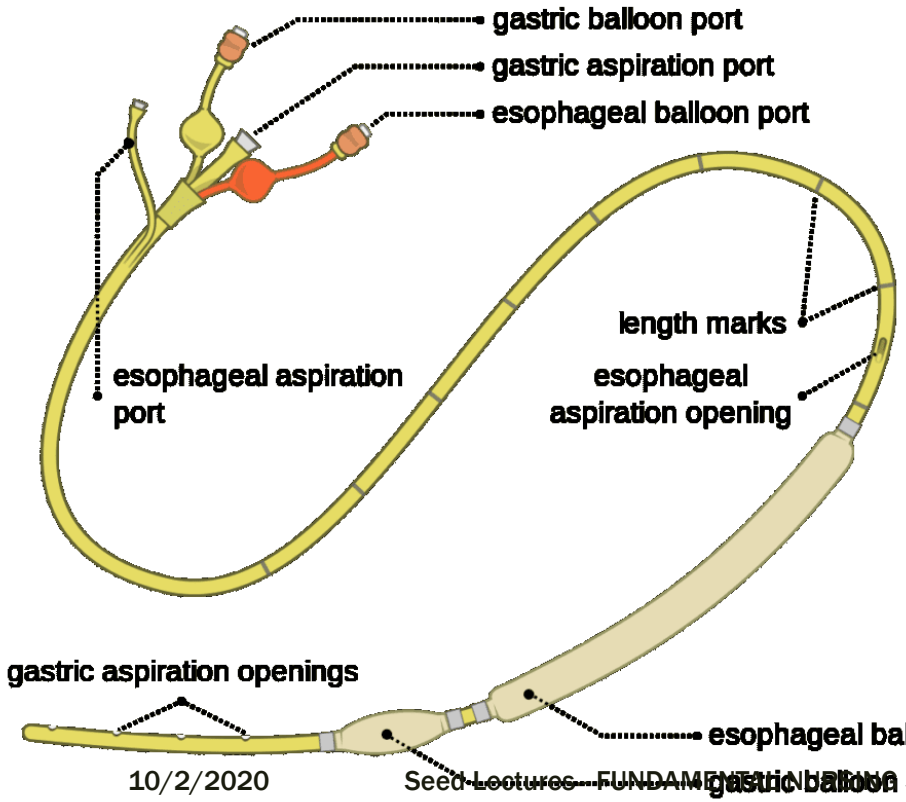

Nasogastric tube
Epiglottis
Trachea
Esophagus
Stomach

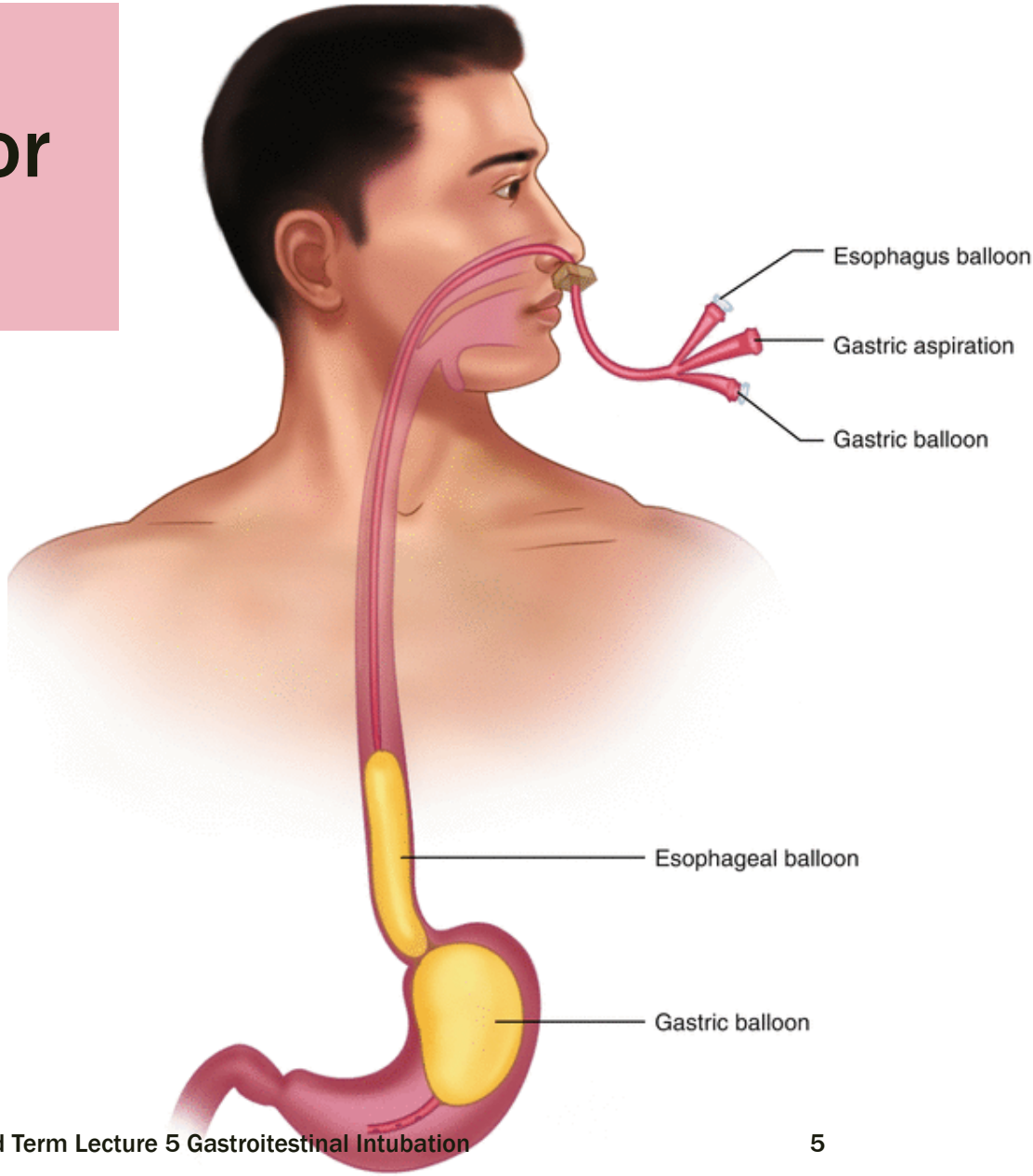

Common Gastrointestinal (GI) Suction Tubes
Single lumen for suction
Large lumen for suctioning of gastric contents
Air vent
LEVIN TUBE
SALEM SUMP TUBE
nurseslabs

Reasons for Intubation

- Performing a gavage (providing nourishment)
- Administering oral medications that the client cannot swallow
- Obtaining a sample of secretions for diagnostic testing
- Performing a lavage (removing substances from the stomach, typically poisons)
- Promoting decompression (removing gas and liquid contents from the stomach or bowel)
- Controlling gastric bleeding, a process called compression or tamponade (pressure)

Controlling gastric bleeding, compression or tamponade (pressure)

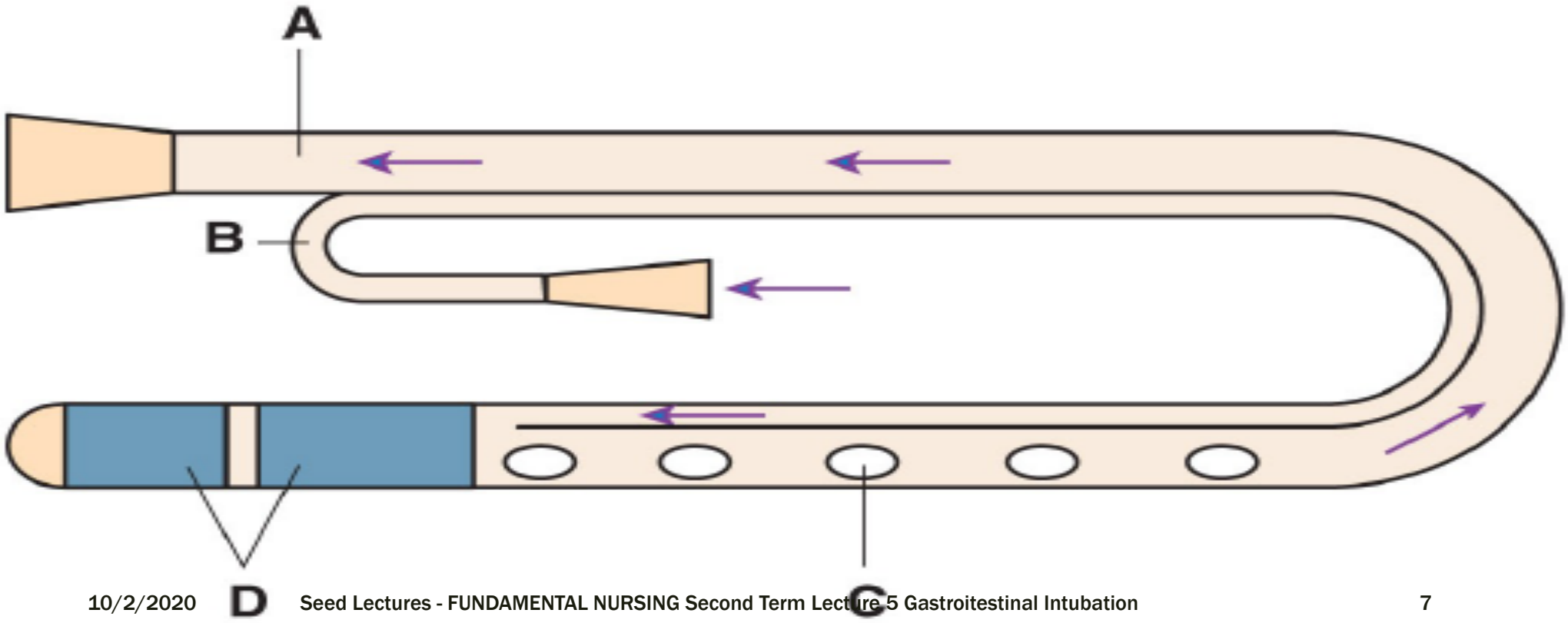




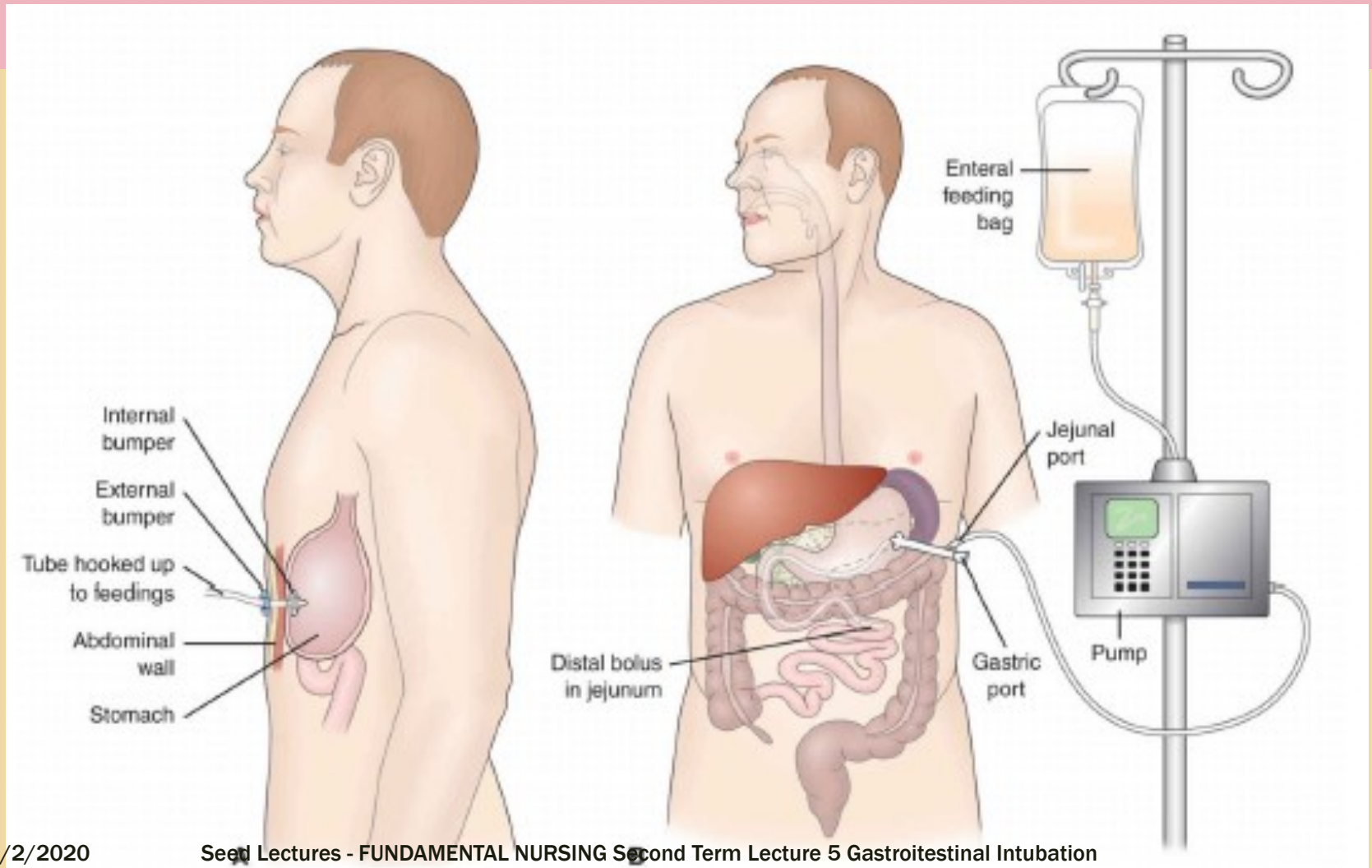
TYPES OF TUBES

- **Orogastric Tubes**
  - ***Such as an Ewald tube, is used in an emergency to remove toxic substances that have been ingested.***
- **Nasogastric tubes**
  - **A Levin tube is a commonly used, single-lumen gastric tube with multiple uses, one of which is decompression.**
  - **Gastric sump tubes (double-lumen tubes) are used almost exclusively to remove fluid and gas from the stomach**
- **Nasointestinal Tubes**
  - **Feeding Tubes**
  - **Intestinal Decompression Tubes**
- **Transabdominal Tubes**
  - **Gastrostomy tube percutaneous endoscopic gastrostomy (PEG) tube (a transabdominal tube inserted under endoscopic guidance)**
  - **Jejunostomy tube percutaneous endoscopic jejunostomy (PEJ) tube (a tube that is passed through a PEG tube into the jejunum)**

An intestinal decompression tube
(A) including the suction lumen,
(B) the vent lumen,
(C) openings for suction,
(D) and the radiopaque tungsten tip.



Transabdominal tubes. (A) percutaneous endoscopic gastrostomy (PEG) tube. (B) percutaneous endoscopic jejunostomy (PEJ) tube.



NASOGASTRIC TUBE MANAGEMENT

- Nasogastric Tube Insertion
- Client preparation
  - Explaining the procedure and giving instructions on how the client can assist while the tube is being passed may further reduce anxiety. One of the most important ways to support clients is to provide them with some means of control.
  - The nurse can establish a signal, such as the client raising a hand, to indicate the need for a pause during the tube's passage.
- Preintubation Assessment, Before insertion, the nurse conducts a focused assessment that includes the client's:
  - Level of consciousness
  - Weight
  - Bowel sounds
  - Abdominal distention
  - Integrity of nasal and oral mucosa
  - Ability to swallow, cough, and gag
  - Any nausea and vomiting

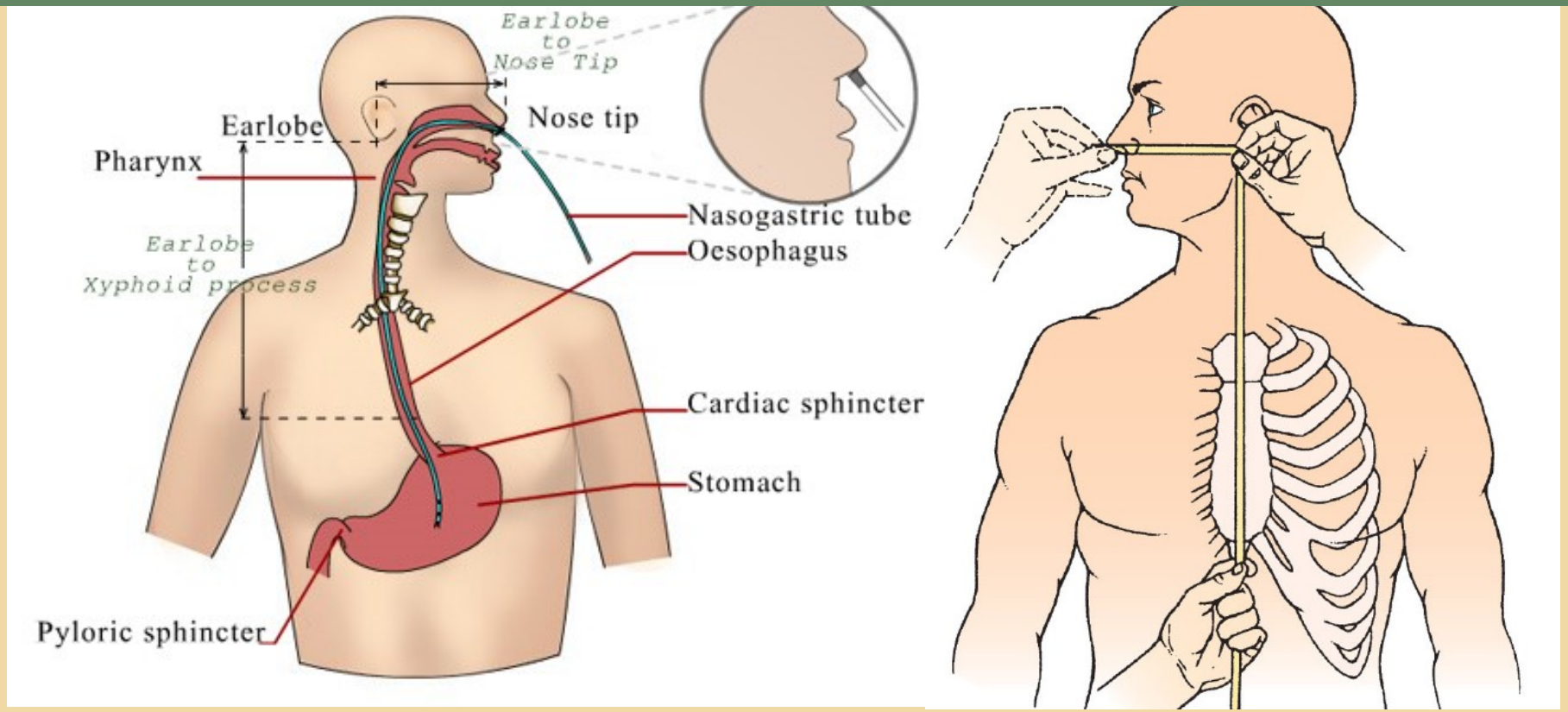
Nasogastric Tube Placement

When inserting a nasogastric tube, the nurse's primary concerns are:

- *to cause as little discomfort as possible,*
- *to preserve the integrity of the nasal tissue, and to*
- *locate the tube within the stomach, not in the respiratory passages.*

Tube measurement

- *Nasal Inspection*
- *Tube measurement (N.E.X. nose – ear – xiphoid)*

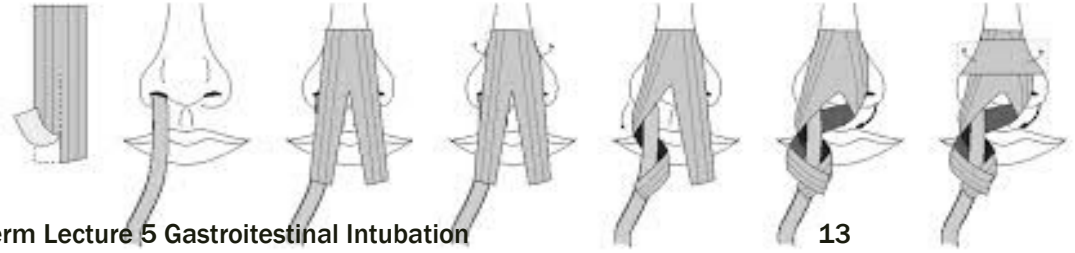


Nasogastric Tube Placement procedure

1. Check physician's order. Explain procedure to patient.
2. Gather equipment.
3. Perform hand hygiene.
4. Don disposable gloves.
5. If nasogastric tube is rubber, place it in a basin with ice for 5–10 minutes or place a plastic tube in a basin of warm water if needed.
6. Assess patient's abdomen.
7. Check nares for patency by asking patient to occlude one nostril and breathe normally through the other. Select nostril through which air passes more easily.
8. Drape chest with bath towel. Have emesis basin and tissues handy.
9. Assist patient to Fowler's position, or 45°.
10. Measure distance to insert the tube by (NEX). Mark tube with marker at E ear level and X xiphoid level.
11. Lubricate tip of tube (at least 1 – 2 inches) with water-soluble lubricant.
12. Ask patient to extend head and insert tube into nostril while directing tube downward and backward. Patient may gag when tube reaches the pharynx the first mark.
13. Encourage him/her to swallow even if no fluids are permitted. Advance tube in a downward-and-backward direction when patient swallows. Stop when patient breathes.
14. Keep advancing tube until second marking is reached. Do not use force. Rotate tube if it meets resistance.
15. Discontinue procedure and remove tube if there are signs of distress, such as gasping, coughing, cyanosis, and inability to speak or hum.

Nasogastric Tube Placement procedure

16. Determine that tube is in patient's stomach.
    - *A. Attach syringe to tube and aspirate a small amount of stomach contents.*
    - *B. Measure pH of paper or a meter.*
    - *C. Visualize aspirated contents checking for color and consistency*
    - *D. Obtain radiograph of placement of tube (as ordered by physician).*
17. Secure tube with tape to patient's nose. Be careful not to pull tube too tightly against nose.
    - *A. Cut a 10 cm piece of tape and split bottom 5cm.*
    - *B. Place unsplit end over bridge of patient's nose.*
    - *C. Wrap split ends under tubing and up and over onto nose.*
18. Attach tube to suction or clamp tube and cap it according to physician's orders.
19. Secure tube to patient's gown by using a rubber band or tape and a safety pin. If double-lumen tube is used, secure vent above stomach level. Attach at shoulder level.
20. Assist or provide patient with oral hygiene at regular intervals.
21. Perform hand hygiene. Remove all equipment and make patient comfortable.
22. Record the insertion skill, type and size of tube. Also document description of gastric contents, which was used, and patient's response.



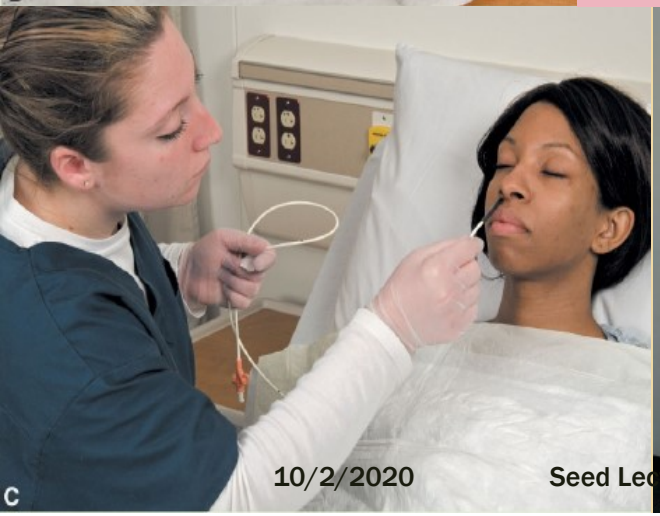
Nasogastric Tube Placement


A

- Wash hands or perform an alcohol-based hand rub
- Put on gloves. After collection of all equipment's including a flexible small-gauge feeding tube with a stylet. Measure the tube (NEX)


B

- Positioning client in semi sitting position with head extended
- Secure stylet if present and lubricate the tip of tube by water lubricant
- Insert tube in nose till the first mark ask client to swallow with sips of water


C


D

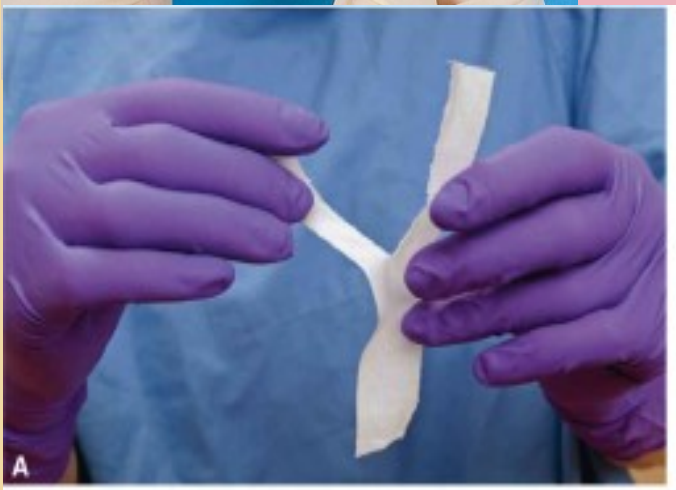
- Continue insertion till reach the second mark with close client observation.
- If client chocked remove tube and reinsert it again.
- After insertion aspirate fluid to check if reached stomach be aspiration of content (yellow or green aspirate)

Nasogastric Tube Placement



- Checking the pH of aspirated fluid.
- Then remove stylet if present.
- Fix the tube well and neatly









Nasogastric tube Placement



First check for the patency of nostrils.

Measure the distance from naris to ear lobule and to the xiphi sternum.

Lubricate the tube .

Pass tube through nostrils.

Ask the patient to swallow water .

Insertion of tube should point towards the xiphoid process.

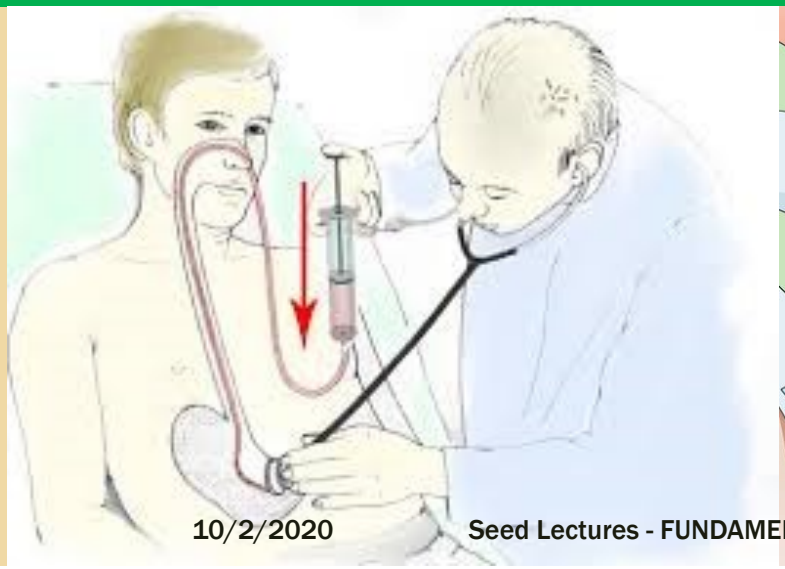
Asses the placement of distal end by air instillation and aspiration

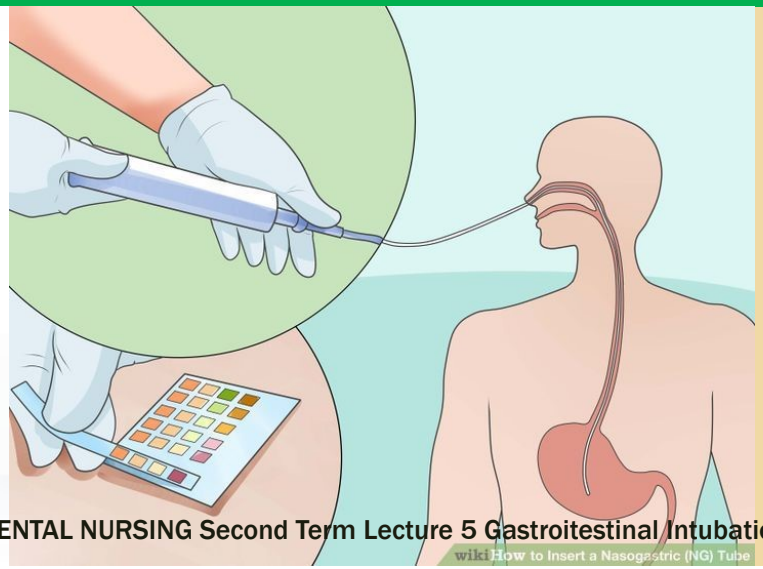
Checking Placement of Intestinal Feeding Tubes

- Before giving any thing by the NGT you must verify that the tube in he stomach by:

1. Aspirate stomach content having yellow or green aspirate
2. Check aspirate on blue litmus paper (Acidity)
3. Use stethoscope to auscultate air flow through 10 ml syringe hearing swashing.
4. X-Ray to check the tip of tube in the stomach.

NB. We could observe flow of tube during insertion by putting the end of tube under water, if there is purples remove and reinsert it again





Use and Maintenance of Nasogastric Tubes

- **Promoting Patency**
  - *Advice pt. having small sipses of water*
  - *Advice client frequent ice chips*
  - *Provide clean water after each feeding*
  - *Check tube frequently*
  - *Give liquid formula with smashed solid food well*
- **Restoring Patency**
  - *Check tube kinking*
  - *Change pt, position*
  - *Rotate the tube several times*
  - *Aspirate stomach content*
  - *Instill 30 ml water to open the tube*
  - *Call physician and change it if patency failed*

TABLE 29-2 Troubleshooting a Poorly Draining Nasogastric Tube

POSSIBLE CAUSES	SOLUTIONS
The drainage holes are adhering to the gastric mucosal wall.	Turn the suction off momentarily. Change the client's position.
The tube is displaced above the esophageal sphincter.	If the measured mark is not at the tip of the nose, remove the tape, advance the tube, check placement, and resecure.
The portable suction machine is disconnected or turned off.	Replace the plug into the electrical outlet or turn on power.
The drainage container is filled beyond capacity.	Empty and record the amount of drainage in the suction container.
The vent is acting as a siphon.	Instill a bolus of air into the vent to restore patency.
The vent is capped or plugged.	Remove the cap and restore the port to atmospheric pressure.
The tubing is kinked or disconnected.	Straighten tubing or reconnect to the suction machine.
The suction is inadequate.	Check that the pressure is 40–60 mmHg.
The cover on the suction container is loose.	Resecure the lid to the container.
A solid particle or thick mucus obstructs the lumen.	Increase suction pressure momentarily. Obtain and implement a medical order for an irrigation.

NG tube feeding procedure

1. Check physician's order, explain procedure to patient and assist to semi-Fowler's position.
2. Gather equipment. Prepare the formula.
3. Perform hand hygiene, Don clean disposable gloves.
4. Aspirate fluid from the feeding tube using a 50-mL syringe, until no more fluid is obtained.
5. Measuring aspirated fluid and instilling it back while containing digested food & stomach juice
6. Instill the formula volume including the volume of residual

Formula = ordered formula volume - residual aspitated volume

1. Flush a feeding tube with 30 mL of water with a large volume syringe before and after an intermittent feeding or administration of medication, any interruption of feeding, and every 4 hours during a continuous feeding.
2. Clients require additional water for hydration and to prevent obstructions within the feeding tube.
3. Document the procedure, and in Intake and output record.

Nasogastric Tube Removal procedure

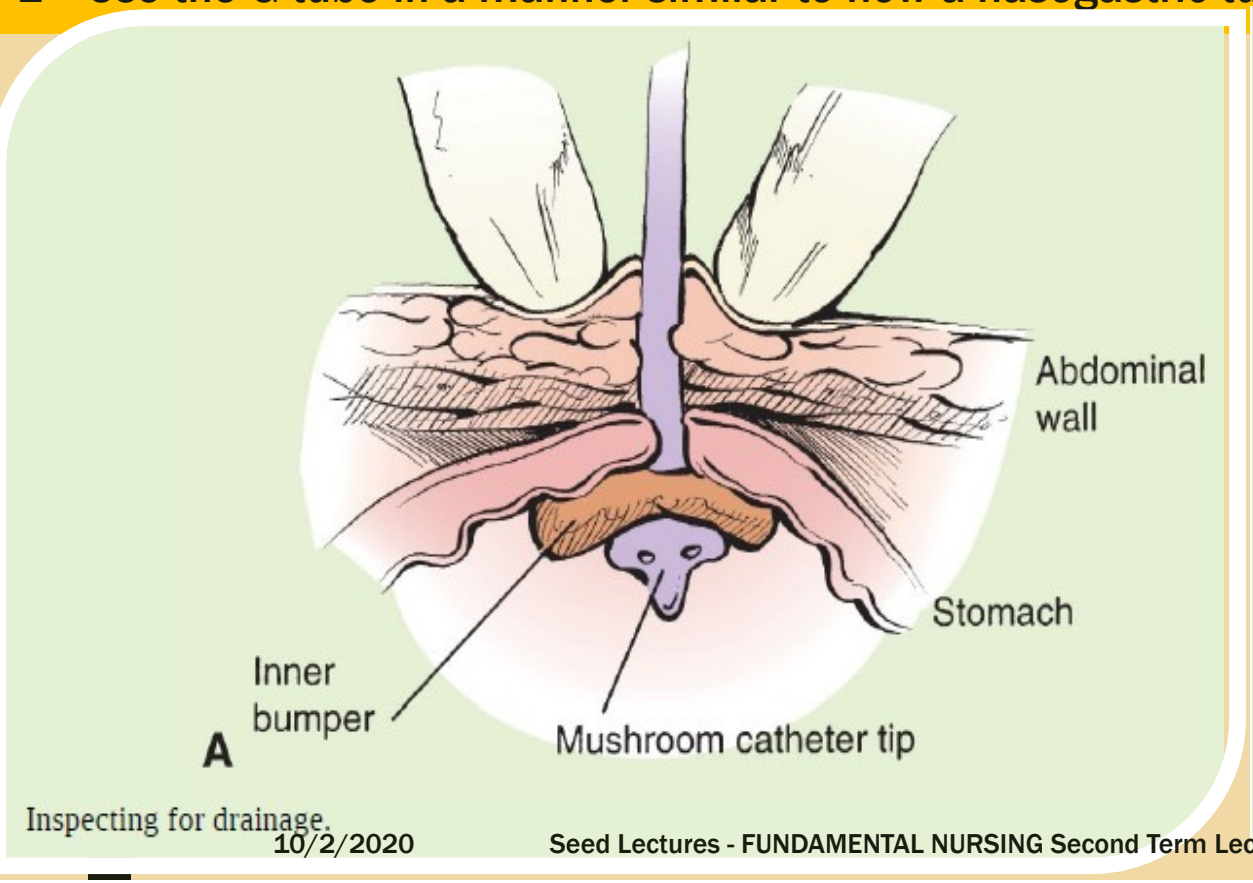
1. Check physician's order for removal of nasogastric tube.
2. Explain procedure to patient and assist to semi-Fowler's position.
3. Gather equipment.
4. Perform hand hygiene.
5. Don clean disposable gloves.
6. Place towel or disposable pad across patient's chest. Give tissues to patient.
7. Unpin tube from and carefully remove adhesive tape from patient's nose.
8. Attach syringe & flush by 10 ml NS solution or clear with 30–50 cc of air.
9. Instruct patient to take a deep breath and hold it.
10. Clamp tube with fingers by doubling tube on itself.
11. Quickly and carefully remove tube while patient holds breath.
12. Place tube in disposable plastic bag. Remove gloves and place in bag.
13. Offer mouth care to patient and facial tissues to blow nose.
14. Measure nasogastric drainage.
15. Remove all equipment and dispose according to agency policy.
16. Perform hand hygiene.
17. Record removal of tube, pt's response, and measure of drainage. Continue to monitor pt. for 2–4 hours after tube removal for distention, nausea, or vomiting.

Managing a Gastrostomy

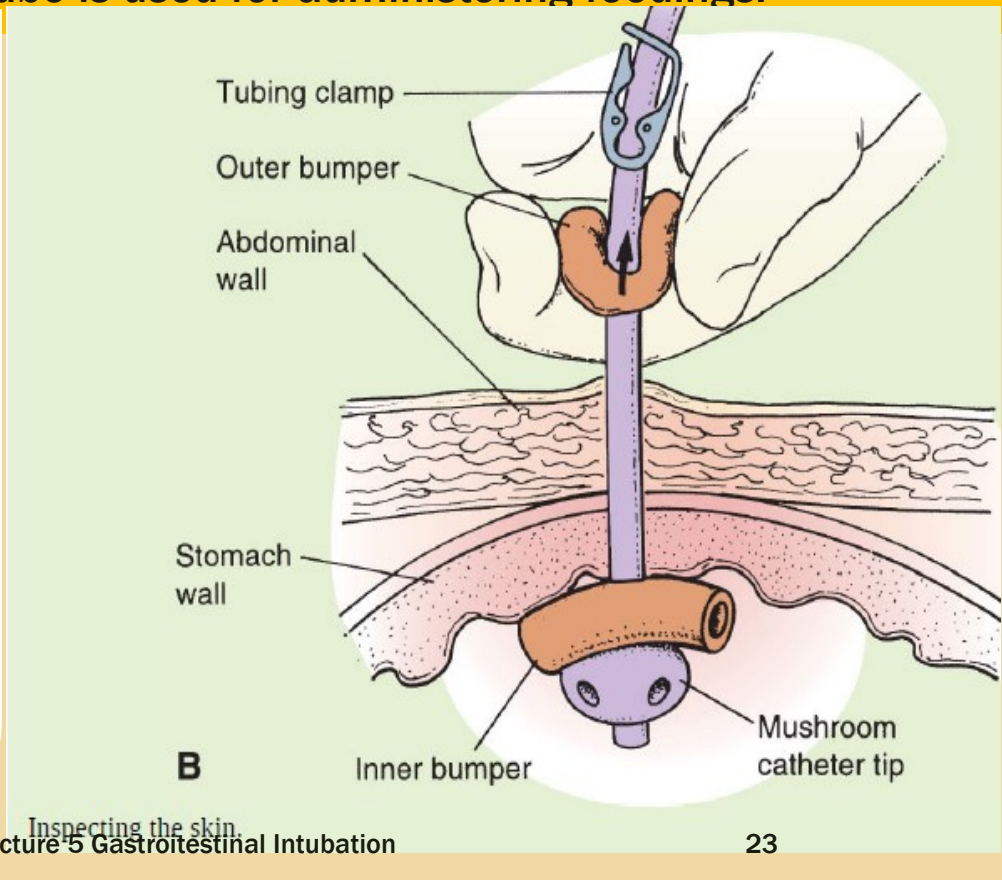
- Wash hands or perform an alcohol-based hand rub, Put on gloves.
- Assess and replace the gauze dressing over a new gastrostomy if it becomes moist.
- Remove dressing after the first 24 hours unless the physician orders otherwise.
- Inspect the skin around the tube daily. Report any redness or tissue maceration
- Make sure that the sutures holding a surgically placed tube are intact.
- Apply a skin barrier ointment such as zinc oxide, hydrocolloid dressing
- Press down on the skin, lift tube about 1 inch (2.5 cm) These steps aid in assessing for drainage.
- Clean the skin with half-strength hydrogen peroxide or 0.9% saline.
- Slide the external bumper down so it is flush with the skin. Sliding re-stabilizes the tube.
- Replace the water in the balloon beneath the bumper weekly using.
- This keeps the balloon fully inflated and prevents tube migration.
- Tape the G-tube to the abdomen or secure it with an abdominal binder.
- Make sure the tube is not kinked and the skin is not stretched.

Managing a Gastrostomy

- Insert a Foley catheter, if the client is not sensitive to latex, 2–5 inches (5–10 cm) within the opening, and inflate the balloon if the tube comes out. *Doing so maintains temporary access to the stomach and, if done within 3 hours of accidental extubation, prevents the site from closing.*
- Use the G-tube in a manner similar to how a nasogastric tube is used for administering feedings.



Inspecting for drainage.



Inspecting the skin.

Causes of Gastrostomy Leaks

- Disconnection between the feeding delivery tube and G-tube.
- Clamped G-tube while tube feeding is infusing.
- Mismatch between the size of the G-tube and stoma.
- Increased abdominal pressure from formula accumulation, sneezing, and coughing.
- Under-inflation of the balloon beneath the skin.
- Less than optimal stoma or stomal location.

Tube feedings

- Tube feedings are used when clients have an intact stomach or intestinal function but are unconscious, have undergone extensive mouth surgery, have difficulty swallowing, or have esophageal or gastric disorders.

Enteral Nutrition

Enteral nutrition (nourishment provided through the stomach or small intestine rather than by the oral route) is delivered by instilling formula through a tube.

Tube-Feeding Schedules

- **Bolus Feedings** (the instillation of liquid nourishment in less than 30 minutes four to six times a day) usually involves 250 to 400 mL of formula per administration.
- **Intermittent Feedings** (the gradual instillation of liquid nourishment four to six times a day) is administered over 30 to 60 minutes, the time most people spend eating a meal. The usual volume is 250 to 400 mL per administration.
- **Cyclic Feedings** (the continuous instillation of liquid nourishment for 8 to 12 hours) is followed by a 16- to 12-hour pause.
- Continuous Feedings (the instillation of liquid nutrition without interruption) is administered at a rate of approximately 1.5 mL/minute.

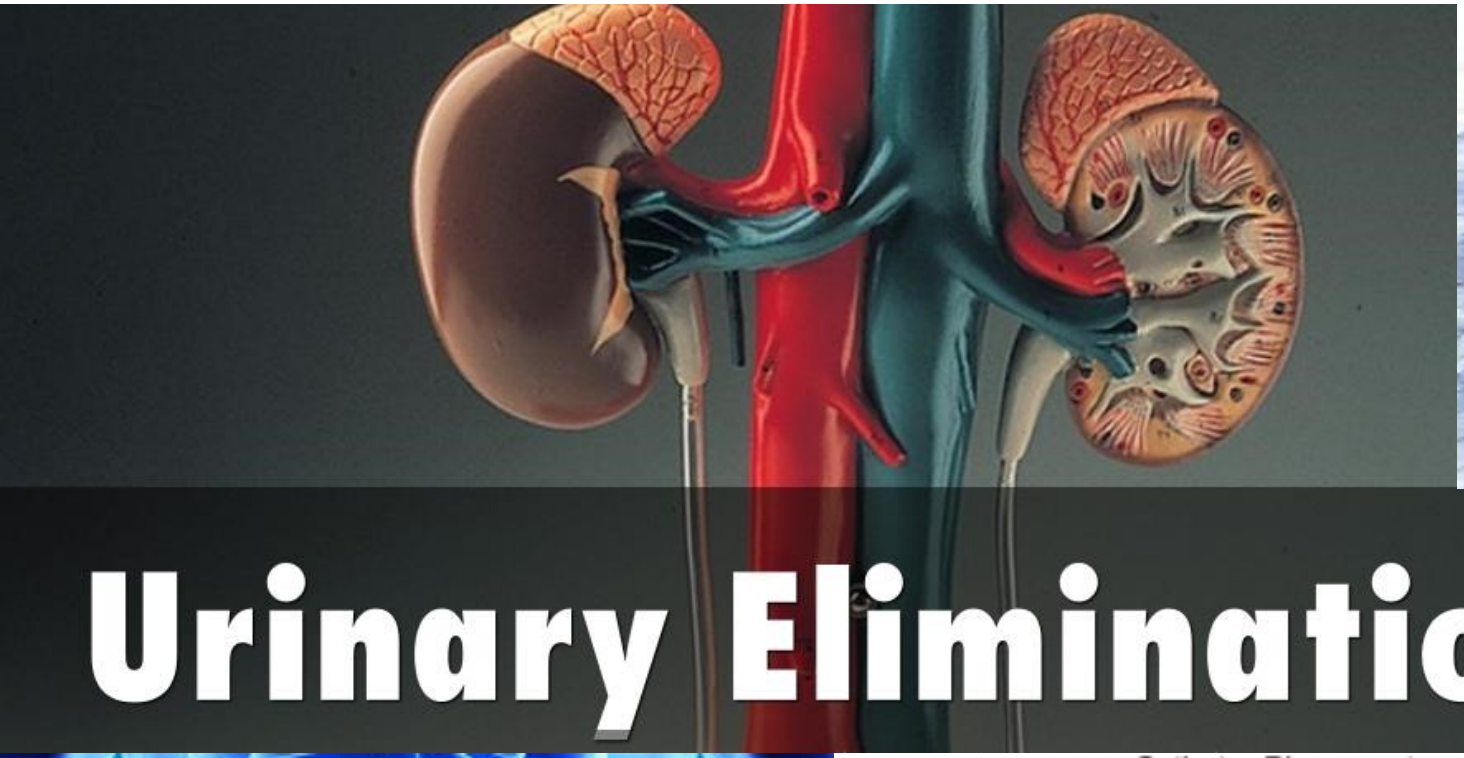
TABLE 29-5 Common Tube-Feeding Problems

PROBLEM	COMMON CAUSES	SOLUTIONS
Diarrhea	Highly concentrated formula Rapid administration Bacterial contamination Lactose intolerance Inadequate protein content Medication side effects	Dilute initial tube feeding to one-quarter to one-half strength. Start at 25 mL/hour and increase rate by 25 mL q12hour. Hang no more than 4 hours' worth of formula. Wash hands. Change formula bag and tubing q24hour. Refrigerate unused formula. Consult with the physician on using a milk-free formula. Raise serum albumin levels with total parenteral nutrition solutions containing supplemental protein, or administer albumin intravenously. Consult with the physician about adjusting drug therapy or administering an antidiarrheal.
Nausea and vomiting	Rapid feeding Overfeeding Air in stomach Medication side effects	Instill bolus and intermittent feedings by gravity. Delay feeding until gastric residual is <100 mL or <20% of hourly volume. Maintain sitting position for at least 30 minutes after feeding. Consult with the physician about ordering medication that facilitates gastric emptying. Administer continuous feedings. Instill feedings within the small intestine. Keep tubing filled with formula or water. Consult with the physician about adjusting drug therapy or administering drugs to control symptoms.
Aspiration	Incorrect tube placement Vomiting	Check placement before instilling liquids. Keep head elevated at least 30 degrees during feedings and for 30 minutes afterward. Keep cuffed tracheostomy and endotracheal tubes inflated. Refer to measures for controlling vomiting.
Constipation	Lack of fiber Dehydration	Change formula. Increase supplemental water. Consult with the physician on giving a laxative, enema, or suppository.
Elevated blood glucose level	Calorie-concentrated formula	Instill diluted formula and gradually increase concentration. Administer insulin according to medical orders.
Weight loss	Inadequate calories	Increase calories in formula. Increase rate or frequency of feedings.
Elevated electrolytes	Dehydration	Increase supplemental water.
Dry oral and nasal mucous membranes	Mouth breathing Dried nasal mucus	Provide frequent oral and nasal hygiene.
Middle ear inflammation	Narrowing or obstruction of eustachian tube from presence of tube in pharynx	Turn from side to side q2hour. Insert a small-diameter feeding tube.
Sore throat	Pressure and irritation from tube	Use a small-diameter feeding tube.
Plugged feeding tube	Instilling crushed or powdered medications through the tube Formula coagulation from drug-food interactions Kinked tube Large molecules in formula	Use liquid medications. Dilute crushed drugs. Flush the tubing liberally after drug administration. Flush tubing with water before and after drug administration. Follow agency policy for alternative flush solutions. Maintain neck in neutral position or change position frequently. Dilute formula. Flush tubing at least q4hour. Use a larger diameter feeding tube.
Dumping syndrome	Rapid and large instillation of highly concentrated formula into the intestine	Administer small, continuous volume. Adjust glucose content of formula.



اللہ یحفظک

God Bless You

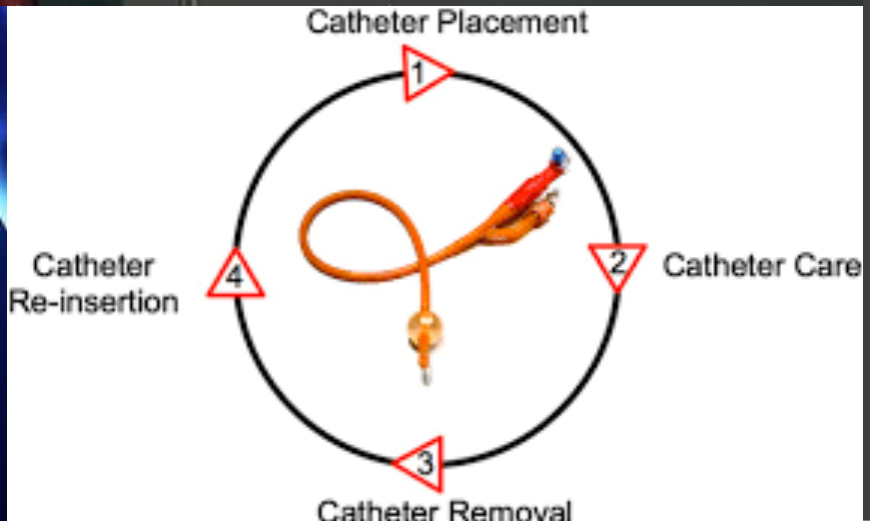




Urinary Elimination

Part I





Learning Objectives

By completion of this chapter, the audience should be able to:

1. *Identify the functions of the urinary system.*
2. *Describe the physical characteristics of urine and factors that affect urination.*
3. *Name four types of urine specimens that nurses commonly collect.*
4. *Identify alternative devices for urinary elimination.*
5. *Define continence training.*
6. *Name three types of urinary catheters.*
7. *Explain why catheter care is important in the nursing management of clients with retention catheters.*
8. *Discuss the purpose for irrigating a catheter and methods for performing this skill.*
9. *Define urinary diversion.*

Urinary Elimination

- ***Urinary elimination*** *(the process of releasing excess fluid and metabolic wastes), or urination, occurs when urine is excreted.*

- *Under normal conditions, the average person eliminates approximately 1,500 to 3,000 mL of urine each day.*

Characteristics of urine

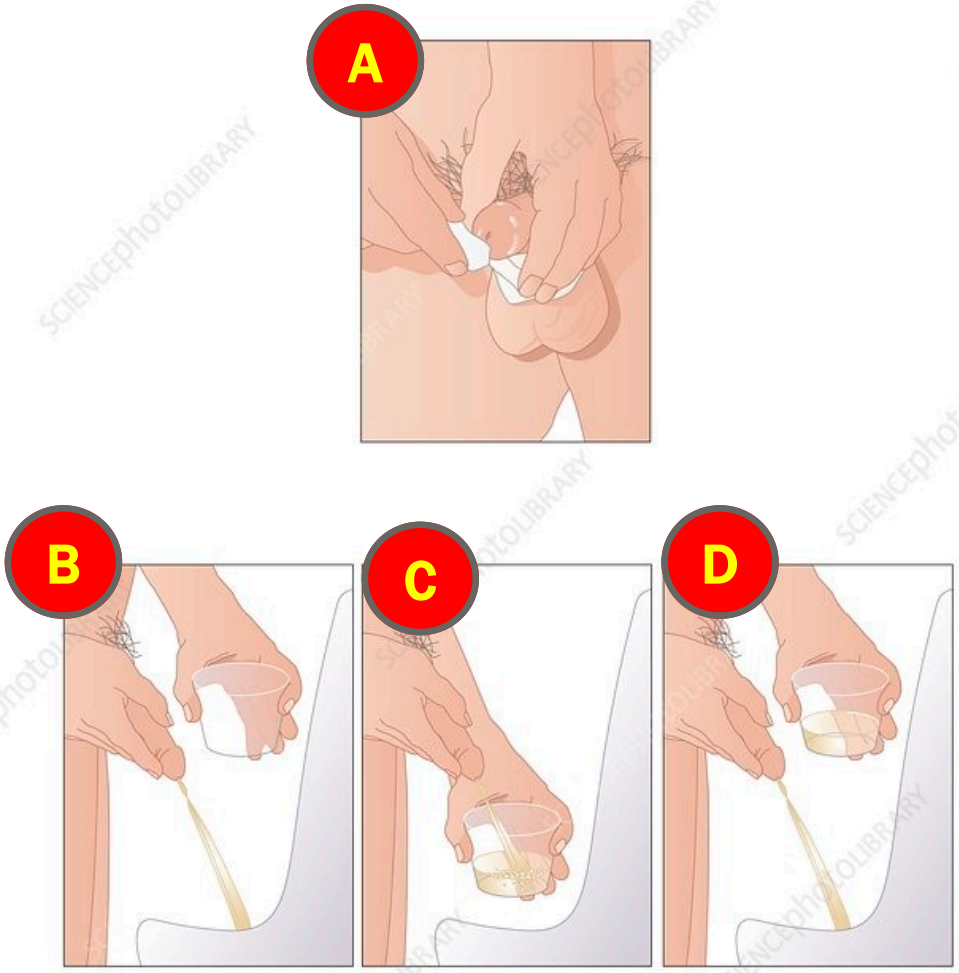
CHARACTERISTIC	NORMAL	ABNORMAL	COMMON CAUSES OF VARIATIONS
Volume	500–3,000 mL/day 1,200 mL/day average	<400 mL/day	Low fluid intake Excess fluid loss Kidney dysfunction
		>3,000 mL/day	High fluid intake Diuretic medication Endocrine diseases
Color	Light yellow	Dark amber	Dehydration
		Brown	Liver/gallbladder disease
		Reddish-brown	Blood
		Orange, green, blue	Water-soluble dyes
Clarity	Transparent	Cloudy	Infection Stasis
Odor	Faintly aromatic	Foul	Infection
		Strong	Dehydration
		Pungent	Certain foods

FACTORS AFFECTING URINARY ELIMINATION

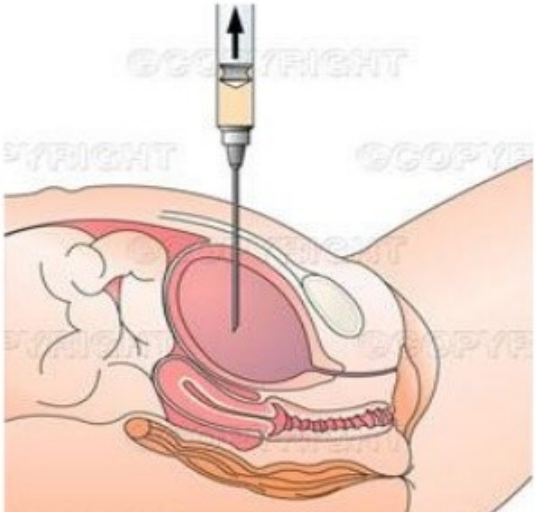
1. *The degree of neuromuscular development and the integrity of the spinal cord.*
2. *The volume of fluid intake and the amount of fluid losses, including those from other sources.*
3. *The amount and type of food consumed.*
4. *The person's circadian rhythm, habits, opportunities for urination, and anxiety.*

Urine Specimen Collection

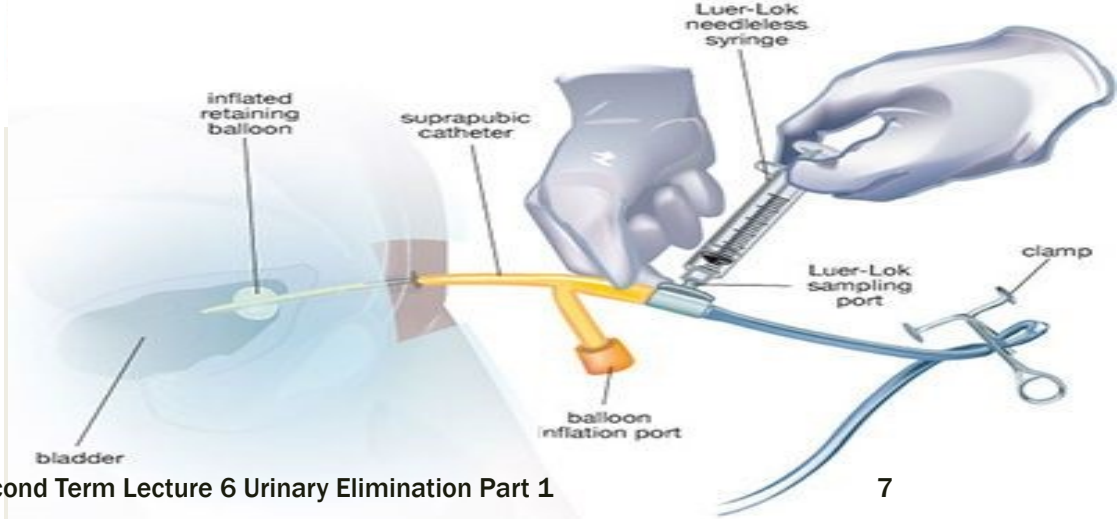
- **Voided Specimens**, is a sample of fresh urine collected in a clean container. The first voided specimen of the day is preferred.
- **Clean-catch specimen** is a voided sample of urine considered sterile and is sometimes called a midstream specimen because of the way it is collected.
- **Catheter Specimens** urine specimen can be collected under sterile conditions using a catheter.
- **24-hour specimen** (a collection of all urine produced in a full 24-hour period) to the laboratory for analysis.



Suprapubic puncture a technically demanding method



- On rare occasions, the health care provider may collect a urine sample by inserting a needle directly into the bladder (suprapubic tap) and draining the urine; this method is used only when a sample is needed quickly and technically competent staff are available



Abnormal Characteristics of Urine

- *Hematuria: urine containing blood*
- *Pyuria: urine containing pus*
- *Proteinuria: urine containing plasma proteins*
- *Albuminuria: urine containing albumin, a plasma protein*
- *Glycosuria: urine containing glucose*
- *Ketonuria: urine containing ketones*

ABNORMAL PATTERNS OF URINARY ELIMINATION

- **Anuria** *means the absence of urine or a volume of* ***100 mL or less in 24 hours****. In urinary suppression, the bladder is empty; therefore, the client feels no urge to urinate.*
  - *This distinguishes anuria from urinary retention, in which the client produces urine but does not release it from the bladder. A sign of* ***urinary retention*** *is a progressively distending bladder.*
- **Oliguria,** *urine output* ***less than 400 mL/24 hours****, indicates the inadequate elimination of urine.*
- **Residual urine,** *or more than* ***50 mL*** *of urine that remains in the bladder after voiding.*
- **Polyuria** *means greater-than-normal urinary elimination and may accompany minor dietary variations. Urine pass* ***more than 3000 ml/day***

ABNORMAL PATTERNS OF URINARY ELIMINATION

- **Nocturia** *(nighttime urination) is unusual because the rate of urine production is normally reduced at night.*
- **Dysuria** *is the difficult or uncomfortable voiding, burning sensation and a common symptom of trauma to the urethra or a bladder infection .*
- **Frequency,** *(the need to urinate often) and.*
- **Urgency** *(a strong feeling that urine must be eliminated quickly) often accompany dysuria.*
- **Incontinence** *means the inability to control either urinary or bowel elimination and is abnormal after a person has achieved earlier continence.*

ASSISTING CLIENTS WITH URINARY ELIMINATION







Providing Continence Training

- Take *data to reveal the client's type of incontinence*.
- Discourage strict limitation of liquid intake.
- Plan a trial schedule for voiding. *This reduces accidental voiding*.
- Plan for voiding every 2 hours during the day and every 4 hours at night.
- Communicate plan with the client, & family.
- Assist the client to a toilet or commode.
- Instruct paralyzed clients to identify any sensation that precedes voiding such as a chill, muscular spasm, restlessness.
- Teach clients with stress incontinence to perform pelvic floor muscle exercises (Kegel exercises).

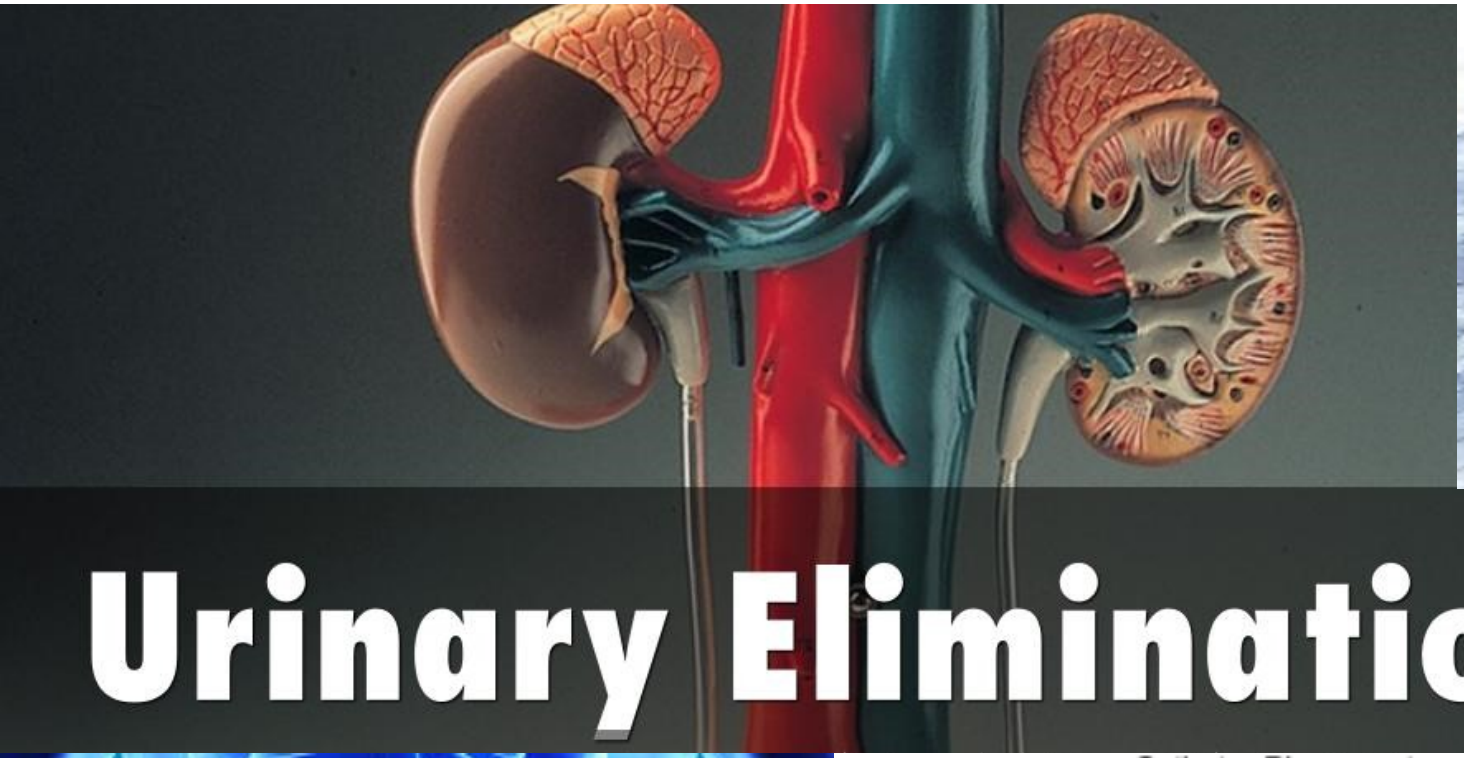
TABLE 30-2 Types of Incontinence

TYPE	DESCRIPTION	EXAMPLE	COMMON CAUSES	NURSING APPROACH
Stress	The loss of small amounts of urine when intra-abdominal pressure rises	Dribbling is associated with sneezing, coughing, lifting, laughing, or rising from a bed or chair.	Loss of perineal and sphincter muscle tone secondary to childbirth, menopausal atrophy, prolapsed uterus, or obesity.	Pelvic floor muscle strengthening Weight reduction
Urge	Need to void perceived frequently, with short-lived ability to sustain control of the flow	Voiding commences when there is a delay in accessing a toilet.	Bladder irritation secondary to infection; loss of bladder tone from recent continuous drainage with an indwelling catheter	Restriction of fluid intake of at least 2,000 mL/day Omit bladder irritants, such as caffeine or alcohol. Administration of diuretics in the morning
Reflex	Spontaneous loss of urine when the bladder is stretched with urine, but without prior perception of a need to void	The person automatically releases urine and cannot control it.	Damage to motor and sensory tracts in the lower spinal cord secondary to trauma, a tumor, or other neurologic conditions	Cutaneous triggering Straight intermittent catheterization
Functional	Control over urination lost because of inaccessibility of a toilet or a compromised ability to use one	Voiding occurs while attempting to overcome barriers such as doorways, transferring from a wheelchair, manipulating clothing, acquiring assistance, or making needs known.	Impaired mobility, impaired cognition, physical restraints, inability to communicate	Clothing modification Access to a toilet, commode, or urinal Assistance to a toilet according to a preplanned schedule
Total	Loss of urine without any identifiable pattern or warning	The person passes urine without any ability or effort to control.	Altered consciousness secondary to a head injury, loss of sphincter tone secondary to prostatectomy, anatomic leak through a urethral/vaginal fistula	Absorbent undergarments External catheter Indwelling catheter
Overflow	Urine leakage because the bladder is not completely emptied; bladder distended with retained urine	The person voids small amounts frequently, or urine leaks around a catheter.	Overstretched bladder or weakened muscle tone secondary to obstruction of the urethra by debris within a catheter, an enlarged prostate, distended bowel, or postoperative bladder spasms	Hydration Adequate bowel elimination Patency of catheter Credé maneuver



Pelvic Floor - Kegel Exercises Personal Training


Thank
You
Obrigado
Merci
Asante
Spasibo
Maake
Kia Ora
Kiitos
Grazie
Matondo
Dank Je
Arigato
Vinaka
Juspaxar
Multumesc
Nirringrazzjak
Ua Tsaug Rau Koj
Mochchakkeram
Terma Kasih
Welalin
Cam on ban
Salamat
Matondo
Chokrane
Dankon
Mamana
Obrigado
Grazie
Kiitos
Asante
Merci
Maake
Spasibo
Kia Ora
Obrigado
Welalin
Kiitos
Chokrane
Matondo
Asante
Grazie
Mochchakkeram
Ua Tsaug Rau Koj
Obrigado
Merci
Spasibo
Kia Ora
Maake
Kiitos
Asante
Grazie
Chokrane
Matondo
Mochchakkeram
Ua Tsaug Rau Koj
Obrigado
Merci
Spasibo
Kia Ora
Maake
Kiitos
Asante
Grazie
Chokrane
Matondo
Mochchakkeram
Ua Tsaug Rau Koj

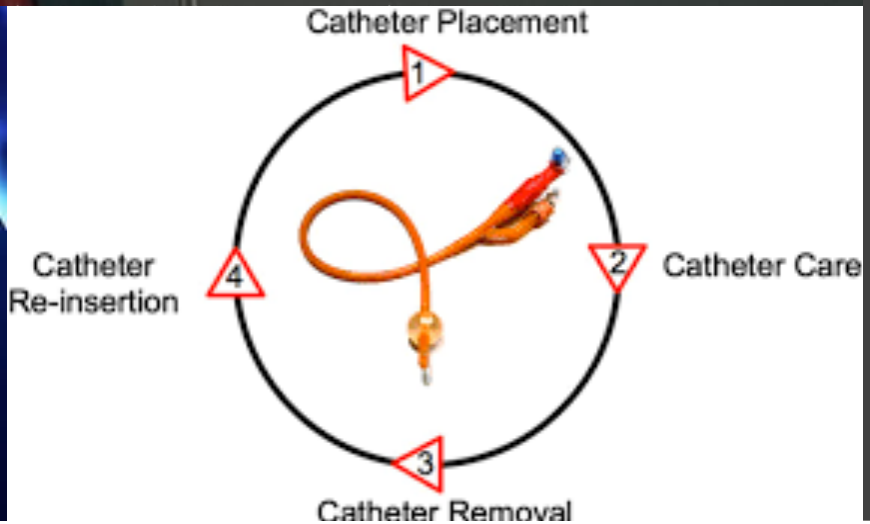




Urinary Elimination

Part II





Managing Retained Urine

- Assist the client to a toilet or commode.
- *Simulate the sound of urination such as by running water from the faucet. Doing so simulates relaxation of the sphincter muscles, allowing the release of urine.*
- Suggest performing **Credé maneuver** (the act of bending forward and applying hand pressure over the bladder.
- *Credé maneuver increases abdominal pressure to overcome the resistance of the internal sphincter muscle*.
- Use **cutaneous triggering** (lightly massaging or tapping the skin above the pubic area). Cutaneous triggering initiates urination in clients who have retained a **voiding reflex** (the spontaneous relaxation of the urinary sphincter in response to physical stimulation).

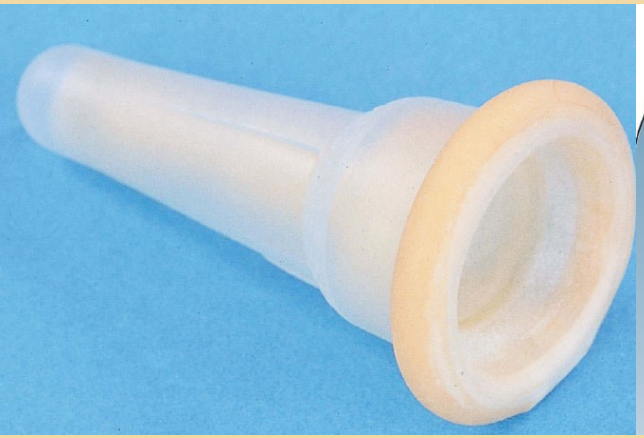


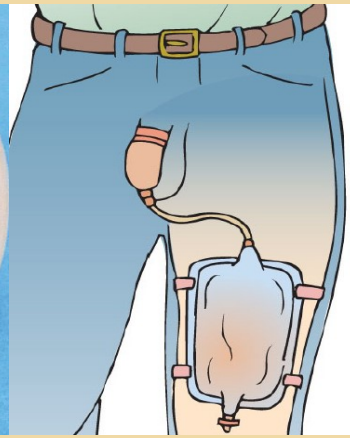
CATHETERIZATION

- Catheterization (the act of applying or inserting a hollow tube), in this case, refers to using a device inside the bladder, or externally about the urinary meatus.
- **A urinary catheter is used for various reasons:**
  1. Keeping incontinent clients dry
  2. Relieving bladder distention when clients cannot void
  3. Assessing fluid balance accurately
  4. Keeping the bladder from becoming distended during procedures such as surgery
  5. Measuring the residual urine
  6. Obtaining sterile urine specimens
  7. Instilling medication within the bladder

Types of Catheters

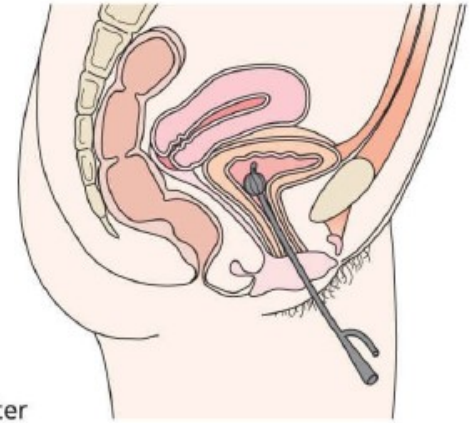
- External Catheters; An external catheter, also known as a condom catheter, (a urine-collecting device applied to the skin for mail clients)

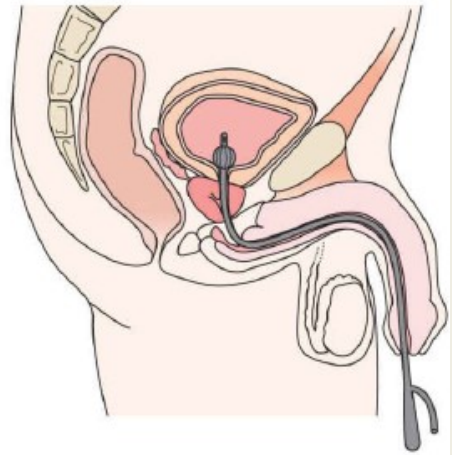




- Straight Catheters; A straight catheter is a urine drainage tube inserted but not left in place. It drains urine temporarily or provides a sterile urine specimen.
- Retention Catheters; also called an indwelling catheter, is left in place for a period of time. The most common type is a Foley catheter.

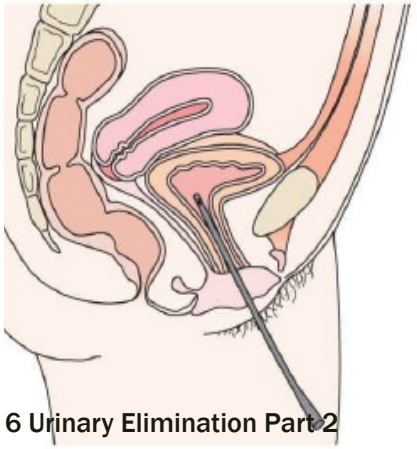


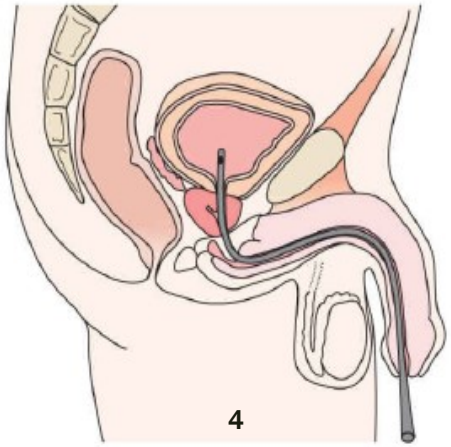




A. Indwelling catheter







B. Straight catheter

Prevent catheter-associated urinary tract infections (CAUTI)

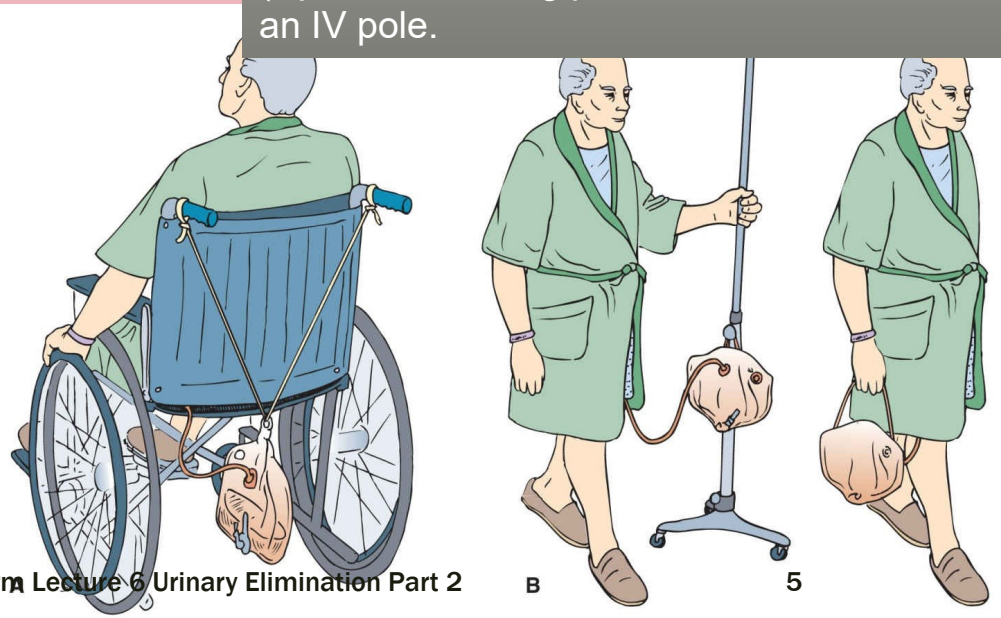
Recommended measures to reduce their incidence:

- Advocate against inappropriate short-term catheter use.
- Secure the catheter appropriately.
- Keep the drainage bag below the level of the bladder at all times.
- Empty the drainage bag regularly.
- Keep the catheter and drainage tube from kinking.
- Maintain a closed drainage system.
- Perform perineal hygiene at least daily.
- Remove urinary catheters in a timely fashion.
- Provide rigorous catheter care.





Techniques for suspending a drainage system below the bladder.
(**A**) A wheelchair patient.
(**B**) An ambulating patient with and without an IV pole.



Removing a Foley Catheter

- Wash your hands or perform an alcohol-based hand rub and put on clean gloves. *These measures follow standard precautions*.
- Empty the balloon by aspirating the fluid with a syringe. *This step ensures that all the fluid has been withdrawn*.
- Gently pull the catheter near the point where it exits from the meatus. *Doing so facilitates withdrawal*.
- Inspect the catheter and discard if it appears to be intact. *This ensures safety*.
- Clean the urinary meatus. *This promotes comfort and hygiene*.
- Monitor the client's voiding especially for the next 8 to 10 hours; measure the volume of each voiding.
- *Findings determine whether or not elimination is normal as well as characteristics of the urine*.

Catheter care

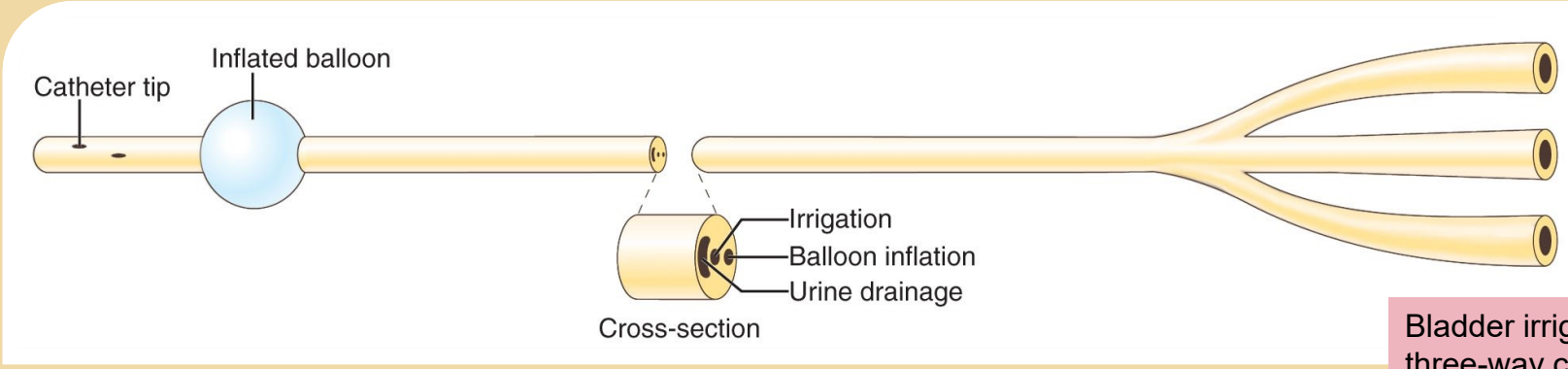
- **Catheter care** (hygiene measures used to keep the meatus and adjacent area of the catheter clean) helps to deter the growth and spread of colonizing pathogens.
- Cleanse the meatus and a nearby section of the catheter at least once a day.
- Gather clean gloves, soap, water, washcloth, towel, and a disposable pad.
- Wash your hands or perform an alcohol-based hand rub.
- Place a disposable pad beneath the hips of a female and beneath the penis of a male.
- Put on clean gloves and wash the meatus, the catheter where it meets the meatus, the genitalia, and the perineum (in that order) with warm, soapy water.
- Remove soiled materials and gloves, and repeat hand hygiene measures. These steps remove colonizing microorganisms.

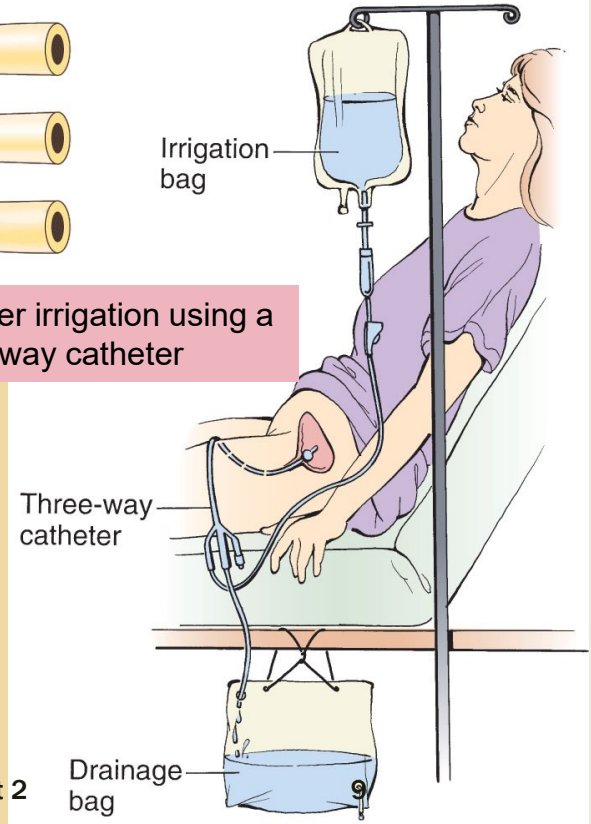
Catheter Irrigation

- **Catheter irrigation** (flushing catheter lumen) is a technique for restoring or maintaining catheter patency.
- **Using an Open System**
  - *An open system is one in which the retention catheter is separated from the drainage tubing to insert the tip of an irrigating syringe.*
  - *Opening the system creates the potential for infection.*
- **Using a Closed System**
  - *A closed system is irrigated without separating the catheter from the drainage tubing,* catheter or drainage tubing must have a self-sealing port.
  - After cleansing the port with an alcohol swab.
  - pierces the port with an 18- or 19-gauge, 1.5 in. needle.
  - *Attach the needle to a 30- to 60-mL syringe containing a sterile irrigation solution.*
  - *The nurse pinches or clamps the tubing beneath the port and instills the solution, then releases the tubing for drainage.*
  - *Record the volume of irrigant as fluid intake or subtracts it from the urine output to maintain an accurate intake and output record.*

Catheter Irrigation

- **Continuous Irrigation**
  - *irrigation (the ongoing instillation of solution) instills irrigation solution into a catheter by gravity over a period of days.*
  - *A three-way catheter is necessary to provide a continuous irrigation.*



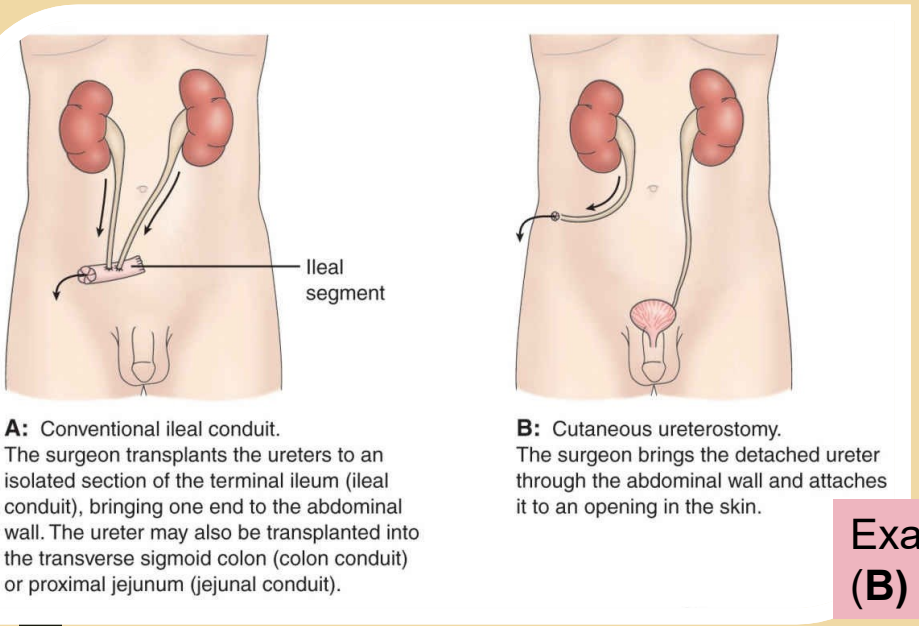


Bladder irrigation using a three-way catheter

- **The steps involved in providing a continuous irrigation are as follows:**
- Hang the sterile irrigating solution from an intravenous pole.
- Purge the air from the tubing.
- Connect the tubing to the catheter port for irrigation.
- Regulate the rate of instillation according to the medical order.
- Monitor the appearance of the urine and volume of urinary drainage.

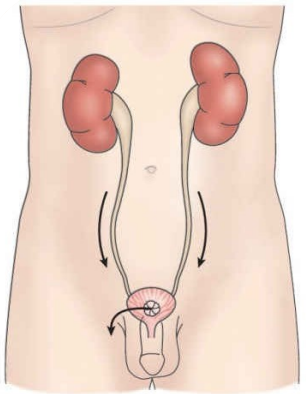
URINARY DIVERSIONS

- In a **urinary diversion**, one or both ureters are surgically implanted elsewhere.
- This procedure is done for various life-threatening conditions. The ureters may be brought to and through the skin of the abdomen or implanted within the bowel (called an ileal conduit).
- A **urostomy** (a urinary diversion that discharges urine from an opening on the abdomen)
- Care for an ostomy, care as for a surgically created opening,

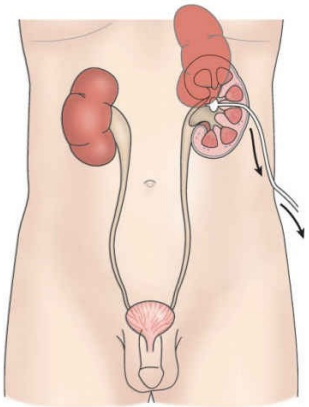


A: Conventional ileal conduit.
The surgeon transplants the ureters to an isolated section of the terminal ileum (ileal conduit), bringing one end to the abdominal wall. The ureter may also be transplanted into the transverse sigmoid colon (colon conduit) or proximal jejunum (jejunal conduit).

B: Cutaneous ureterostomy.
The surgeon brings the detached ureter through the abdominal wall and attaches it to an opening in the skin.



C: Vesicostomy.
The surgeon sutures the bladder to the abdominal wall and creates an opening (stoma) through the abdominal and bladder walls for urinary drainage.



D: Nephrostomy.
The surgeon inserts a catheter into the renal pelvis via an incision into the flank or, by percutaneous catheter placement into the kidney.

Examples of urinary diversions. (**A)** An ileal conduit.
(**B)** A cutaneous ureterostomy. (**C)** Vesicostomy. (**D)** Nephrostomy.


Thank
You
Merci
Obrigado
Asante
Grazie
Maake
Kiitos
Spasibo
Multumesc
Matondo
Mochchakkeram
Dank Je
Arigato
Vinaka
Juspaxar
Mamana
Nirringrazzjak
Ua Tsaug Rau Koj
Terma Kasih
Welalin
Cam on ban
Kia Ora
Salamat
Matur Nuwun
Chokrane
Dankon
Raibh Maith Agat
Obrigado
Asante
Grazie
Kiitos
Merci
Matondo
Mochchakkeram
Ua Tsaug Rau Koj
Chokrane
Welalin
Multumesc
Spasibo
Nirringrazzjak
Juspaxar
Mamana
Dankon
Kia Ora
Salamat
Matur Nuwun
Chokrane
Raibh Maith Agat
Ua Tsaug Rau Koj
Mochchakkeram

BOWEL ELIMINATION

Bowel elimination

- **Defecation** (bowel elimination) is the act of expelling **feces** (stool) from the body.
- **Peristalsis** means the rhythmic contractions of intestinal smooth muscle that facilitate defecation.
- **Gastrocolic reflex** or **gastrocolic** response is a physiological **reflex** that controls the motility, or peristalsis, of the gastrointestinal tract following a meal.
- **Valsalva maneuver** (closing the glottis and contracting the pelvic and abdominal muscles to increase abdominal pressure) facilitates thedefication process.



FACTOR	EFFECT
Types of food consumed	Influence color, odor, volume, and consistency of stool, and fecal velocity
Fluid intake	Influences moisture content of stool
Drugs	Slow or speed motility
Emotions	Alter bowel motility
Neuromuscular function	Affects the ability to control rectal muscles
Abdominal muscle tone	Affects the ability to increase intra-abdominal pressure (Valsalva maneuver)
Opportunity for defecation	Inhibits or facilitates elimination

Elimination Patterns

1. adults should identify their own patterns of bowel regularity, which can range from three times a day to three times a week.
2. include daily exercise.
3. eat high fiber foods on a regular basis.
4. drink 8 to 10 glasses of liquid a day (unless contraindicated).
5. respond to the urge to defecate as soon as possible.







Characteristics of Stool



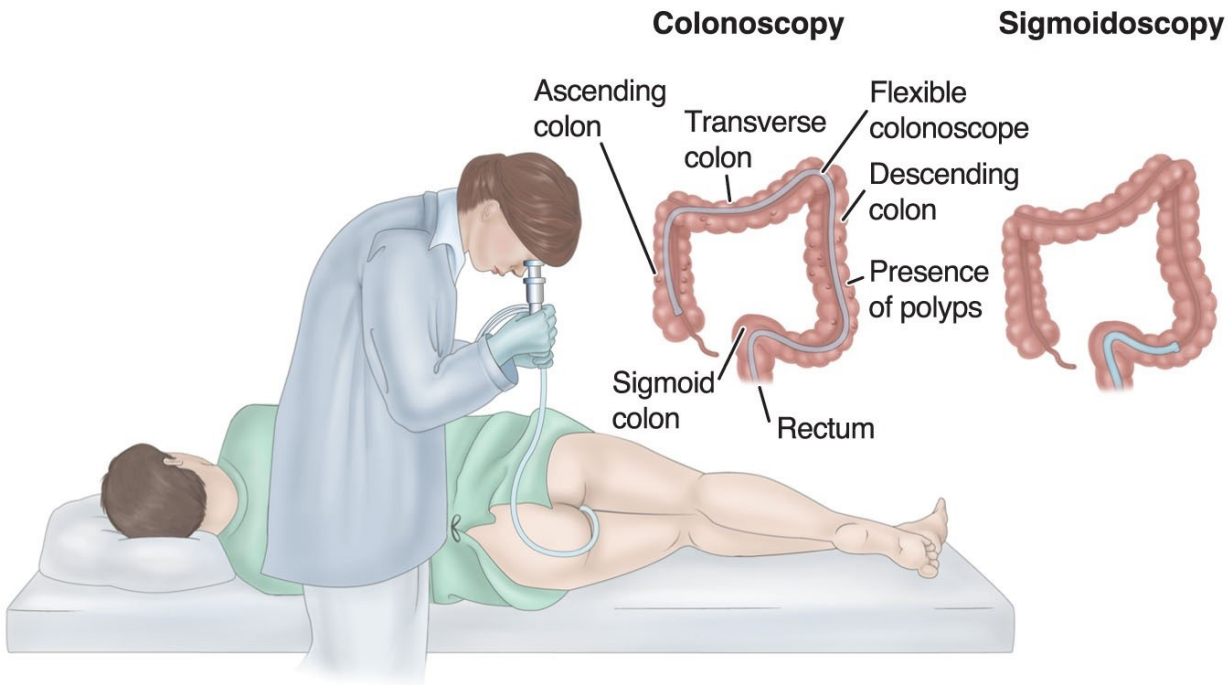
CHARACTERISTIC	NORMAL	ABNORMAL
Color	Brown	Black
		Clay colored (tan)
		Yellow
		Green
)dor	Aromatic	Foul
:onsistency	Soft, formed	Soft, bulky
		Hard, dry
		Watery
		Paste like
hape	Round, full	Unformed
		Flat
		Pencil-shaped
		Stone like
Components	Undigested fiber	Worms
		Blood
		Pus
		Mucus

Testing for Colorectal Disorders

- *The incidence of colorectal cancer increases with age; it is the third most common cancer diagnosed in men and women.*
- *One early sign is a change in bowel elimination patterns and stool characteristics. Blood in stool* (**melena**) *is an abnormal characteristic that may be invisible to the naked eye.*
- **Fecal Occult Blood Test (FOBT)**, a self-collected screening test from three separate stools, may be obtained by clients to detect heme, within stool as it passes through the small and large intestine.
- **Fecal Immunochemical Test (FIT)** is the preferred alternative to the FOBT. The FIT is more specific than an FOBT because it uses antibodies to detect **globin**, a protein removed from heme, which is present exclusively in the lower intestine.
- **Stool DNA Test,** is now available to detect mutant DNA from tumor cells present in stool.

Endoscopic Examinations

- **Colonoscopy,** *examinations, a visual inspection of the interior colon using a flexible lighted endoscope, which is considered the most accurate test for detecting colorectal cancer.*
- **Sigmoidoscopy,** *visual endoscopic inspection limited to the sigmoid* portion of the large intestine, every 5 to 10 years.
- **Colonography,** *a CT scan without a colonoscope, sometimes referred to as a virtual colonoscopy, every 5 years.*



COMMON ALTERATIONS IN BOWEL ELIMINATION

- **Constipation** *is an elimination problem characterized by dry, hard stool that is* difficult to pass..
- *Various accompanying signs and symptoms include the following:*
  - *Complaints of abdominal fullness or bloating*
  - *Abdominal distention*
  - *Complaints of rectal fullness or pressure*
  - *Pain on defecation*
  - *Decreased frequency of bowel movements*
  - *Inability to pass stool*
  - *Changes in stool characteristics such as oozing, liquid stool or hard, small stool*

Constipation is classified into one of four distinct types (primary, secondary, iatrogenic, and pseudoconstipation), according to the underlying cause.

Classifications of Constipation

- **Primary Constipation**
  - *Primary or simple constipation is well within the treatment domain of nurses. It results from lifestyle factors such as inactivity, inadequate intake of fiber, insufficient fluid intake, or ignoring the urge to defecate.*
- **Secondary Constipation**
  - *Secondary constipation is a consequence of a pathologic disorder such as a partial bowel obstruction. It usually resolves when the primary cause is treated.*
- **Iatrogenic Constipation**
  - *Iatrogenic constipation occurs as a consequence of other medical treatments. For example, prolonged use of narcotic analgesia tends to cause constipation.*
- **PseudoConstipation**
  - *is a term used when clients believe themselves to be constipated, even though they are not..*

REMEMBER

Many individuals believe that a daily bowel movement is the norm

and turn to laxative use to promote a daily bowel movement.

If a client reports chronic constipation, investigate the client's use of laxatives.

Laxatives, like many medications, can lead to dependence with long-term use.

Fecal impaction

- Fecal impaction occurs when a large, hardened mass of stool interferes with defecation, making it impossible for the client to pass feces voluntarily.
- **Removing a Fecal Impaction**
  1. *Wash your hands or perform an alcohol-based hand rub. Put on clean examination gloves.*
  2. *Provide privacy. Privacy demonstrates respect for the client's dignity.*
  3. *Place the client in a Sims' position. This position facilitates access to the rectum.*
  4. *Cover the client with a drape and place a disposable pad under the client's hips.*
  5. *Place a bedpan conveniently on the bed. The bedpan acts as a container for removed stool.*
  6. *Lubricate the forefinger of your dominant gloved hand. Lubrication eases insertion within rectum.*
  7. *Insert your lubricated finger within the rectum to the level of the hardened mass.*
  8. *Move your finger about slowly and carefully to break up the mass of stool.*

Removing a Fecal Impaction cnt.

9. *Withdraw segments of the stool and deposit them in the bedpan. Removal reduces the internal mass of stool.*
10. *Provide periods of rest but continue until the mass has been removed or sufficiently reduced. Doing so restores patency to the lower bowel.*
11. *Clean the client's rectal area; dispose of the stool and soiled gloves; repeat hand hygiene measures.*


A


B

REMEMBER

A.Client in a Sims' position

B.Stool is removed digitally

Flatulence

- **Flatulence** or **flatus** (an excessive accumulation of intestinal gas) results from:
  - *Swallowing air while eating or from sluggish peristalsis.*
  - *Another cause is the gas that forms as a by-product of bacterial fermentation in the bowel.*
  - *Vegetables such as cabbage, cucumbers, and onions are commonly known for producing gas.*
  - *Beans are other gas formers.*

When clients are extremely uncomfortable and ambulating does not eliminate flatus, the nurse may insert a rectal tube to help the gas escape.

Diarrhea

- **Diarrhea** is the urgent passage of watery stool and commonly is accompanied by abdominal cramping:
  - *Simple diarrhea usually begins suddenly and lasts for a short period.*
  - *Other associated signs and symptoms include nausea, vomiting and blood or mucus in the stools.*
  - *Usually, diarrhea is a means of eliminating an irritating substance such as tainted food or intestinal pathogens.*
  - *Diarrhea may also result from emotional stress, dietary indiscretions, laxative abuse, or bowel disorders.*

Resting the bowel temporarily may relieve simple diarrhea, antidiarrheal drugs are not recommended.

Fluid replacement is the best choice in management of diarrhea

Fecal Incontinence

- **Fecal incontinence** is the inability to control the elimination of stool. It does not necessarily imply that stool is loose or watery.
  - *In some instances, bowel function is normal, but incontinence results from neurologic changes that impair muscle activity, sensation, or thought processes.*
  - *Even a fecal impaction may be an underlying cause of incontinence.*
  - *Incontinence also may occur when a person cannot reach a toilet in time to eliminate, such as after taking a harsh laxative.*

Chronic fecal incontinence can be devastating socially and emotionally.

Client and Family Teaching

- The nurse teaches the client and the family as follows:
- Eat regularly and nutritiously.
- Monitor the pattern of incontinence to determine if it occurs at a similar time each day.
- Sit on the toilet or bedside commode before the time elimination tends to occur.
- Consult the physician about administering a suppository or an enema every 2 to 3 days to establish a pattern for bowel elimination.
- Use moisture-proof undergarments and absorbent pads to protect clothing and bed linen.
- Teach caregivers to do the following:
  - *Do not imply, verbally or nonverbally, that the client is to blame for the incontinence or that cleaning him or her is disgusting.*
  - *Avoid anything that connotes diapering, to preserve the client's dignity and self-esteem.*

MEASURES TO PROMOTE BOWEL ELIMINATION

- **Inserting a Rectal Suppository**
  - *A **suppository** (an oval or cone-shaped mass that melts at body temperature) is inserted into a body cavity such as the rectum. The most common reason for inserting a suppository is:*
  - *to deliver a drug that will promote the expulsion of feces.*
  - *Other medications, such as drugs to control vomiting and reduce fever,*
- **Administering an Enema**
- An **enema** introduces a solution into the rectum. Nurses give enemas to:
  1. *Cleanse the lower bowel (most common reason).*
  2. *Soften feces.*
  3. *Expel flatus.*
  4. *Soothe irritated mucous membranes.*
  5. *Outline the colon during diagnostic X-rays.*
  6. *Treat worm and parasite infestations.*

Cleansing Enemas

Cleansing enemas use different types of solution to remove feces from the rectum. Defecation usually occurs within 5 to 15 minutes after administration.

SOLUTION	AMOUNT (mL)	MECHANISM OF ACTION
Tap water	500–1,000	Distends rectum, moistens stool
Normal saline	500–1,000	Distends rectum, moistens stool
Soap and water	500–1,000	Distends rectum, moistens stool, irritates local tissue
Hypertonic saline	120	Irritates local tissue and draws water into the bowel
Mineral, olive, or cottonseed oil	120–180	Lubricates and softens stool





Enemas may be administered by (A) using a commercially prepared solution or with (B) a disposable enema bag and individually prepared solution.

Retention Enemas

- A **retention enema** uses a solution held within the large intestine for a specified period, usually at least 30 minutes. Some retention enemas are not expelled at all.
  - *One type of retention enema is called an oil retention enema because the fluid instilled is mineral, cottonseed, or olive oil.*
  - *Oils lubricate and soften the stool, so it can be expelled more easily.*

Administering a Hypertonic Enema Solution

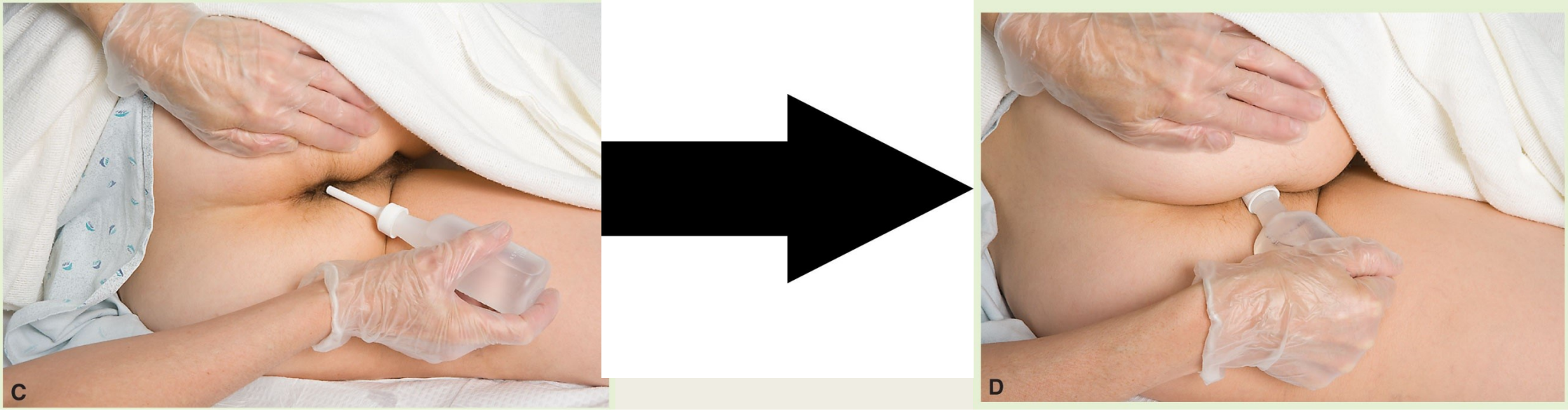
- Warm the container of solution (if it is cold) by placing it in a basin or sink of warm water.
- Assist the client to a Sims position or use a knee–chest position.
- Wash hands or use an alcohol-based hand rub and put on gloves.
- Remove the cover from the lubricated tip & Cover the tip with additional lubricant.
- Invert the container and compress the fluid toward the enema tip





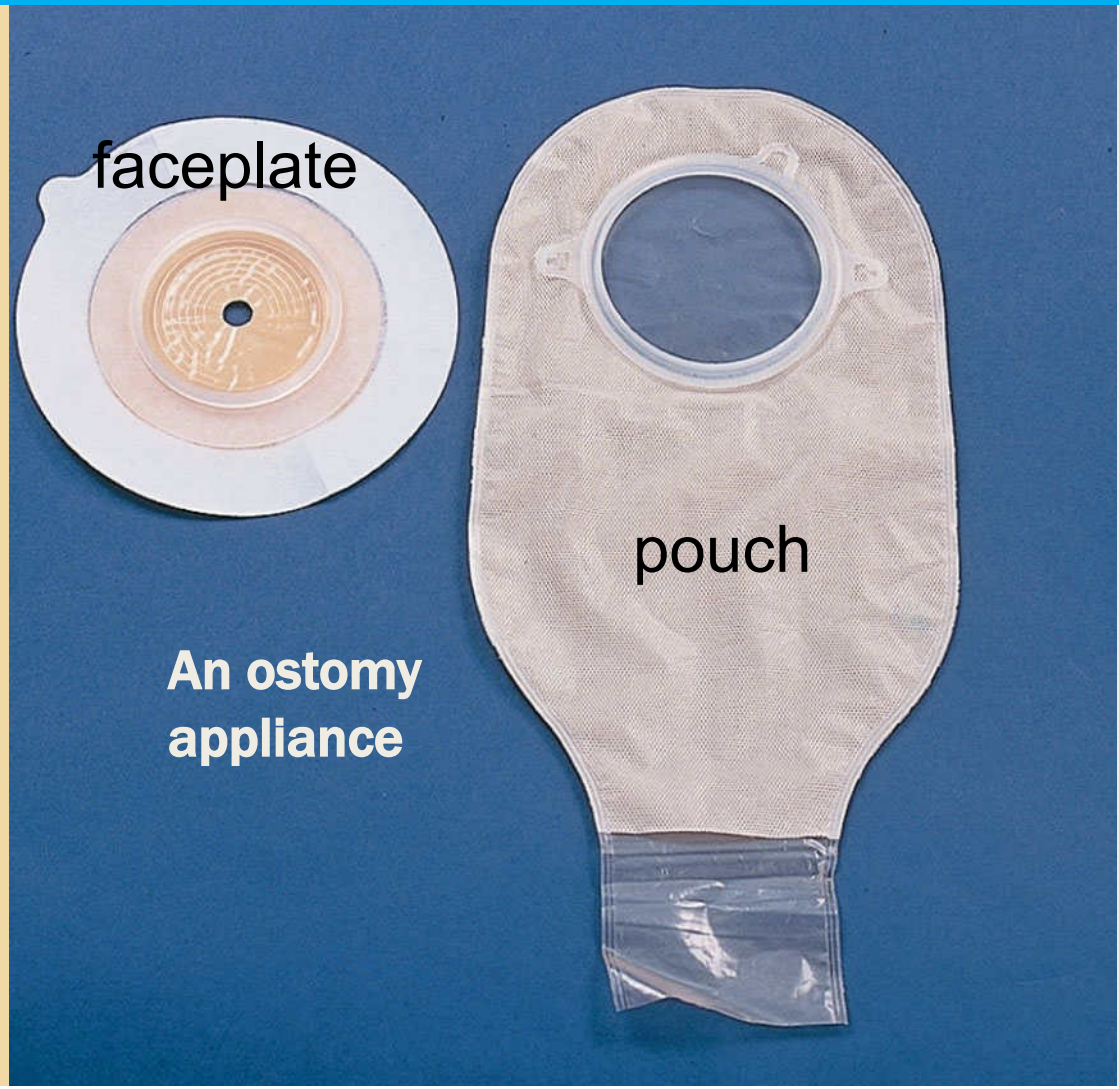
Administering a Hypertonic Enema Solution cont.

- Insert the full length of the tip within the rectum
- Apply gentle, steady pressure 1 to 2 minutes or until the solution has been completely administered.
- Encourage the client to retain the solution for 5 to 15 minutes.
- Clean the client and position for comfort.
- Discard the container, remove gloves, and perform hand hygiene measures.
- Monitor the client and document the procedure and results.



OSTOMY CARE

- A client with an ostomy (a surgically created opening to the bowel or other structure) requires additional care for promoting bowel elimination.
- Two examples of intestinal ostomies are an
  - *ileostomy (a surgically created opening to the ileum) and*
  - *colostomy (a surgically created opening to a portion of the colon*



Applying an Ostomy Appliance

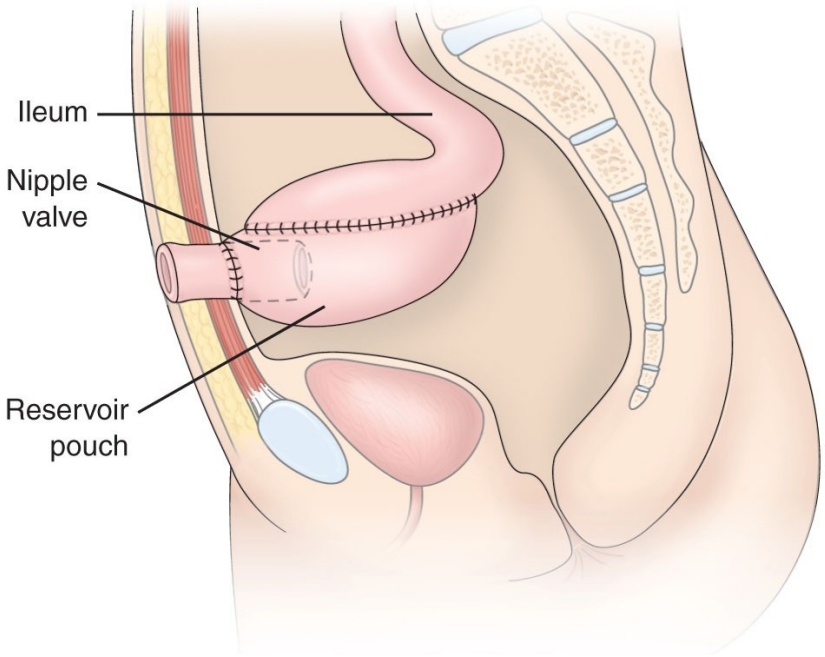
- Various appliances are available, but all consist of:
  - *pouch for collecting stool and a*
  - *faceplate, or disk, that attaches to the abdomen.*
- The stoma protrudes through an opening in the center of the appliance.
- The pouch fastens into position when pressed over the circular support on the faceplate.
- Some clients prefer a type that also fastens to an elastic belt worn around the waist.
- The belt helps to support the weight of the fecal material and prevents the faceplate from being pulled away from the abdomen.
- The client empties the pouch by releasing the clamp at the bottom.



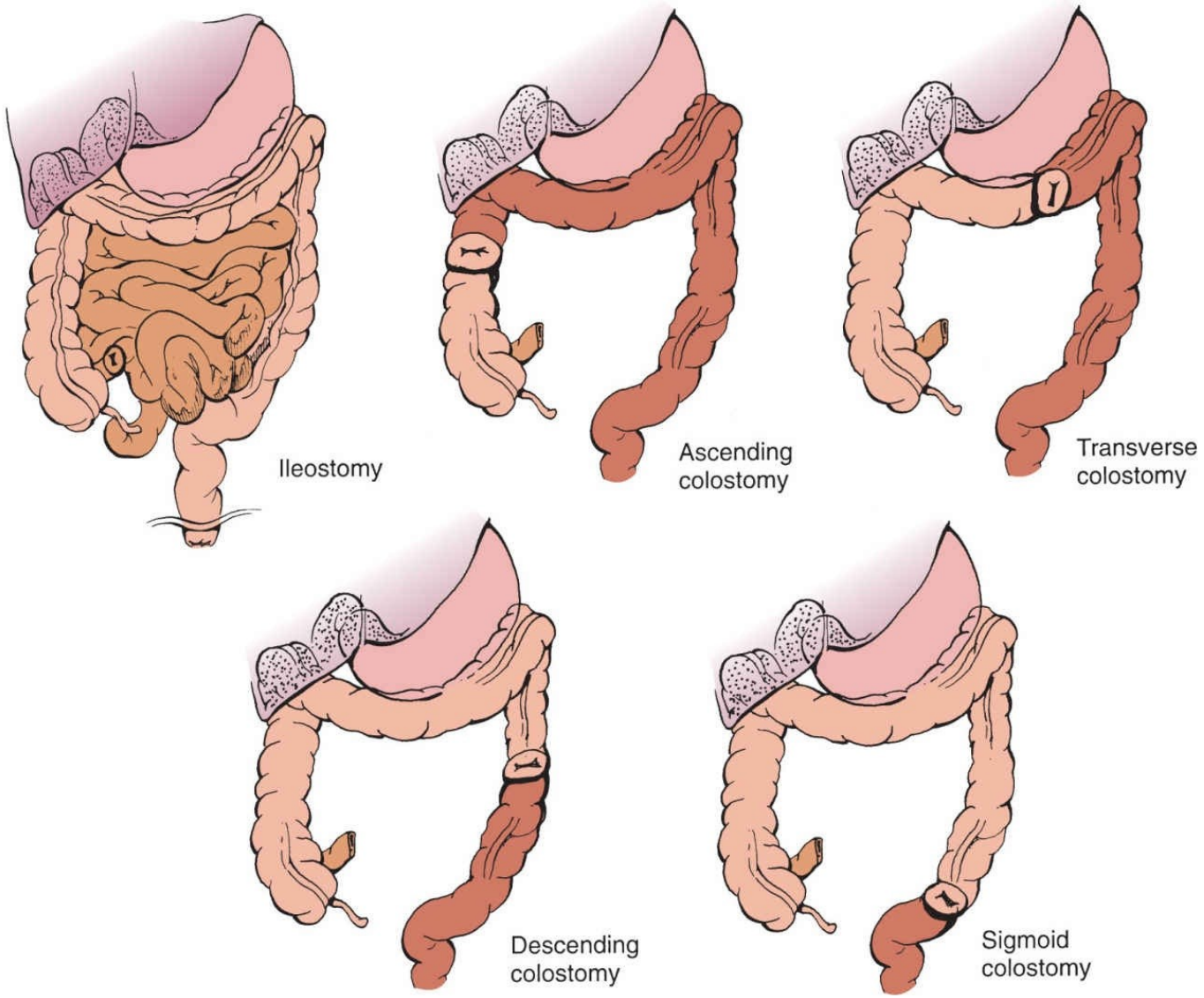


A. Remove the face plate	D. Attach the pouch	G. Attaching the appliance to the skin
B. Measuring the stoma	E. Sealing the pouch	
C. Trimming the stomal opening	F. Removing the adhesive backing	

The locations of intestinal ostomies



The locations of intestinal ostomies



Providing Peristomal Care

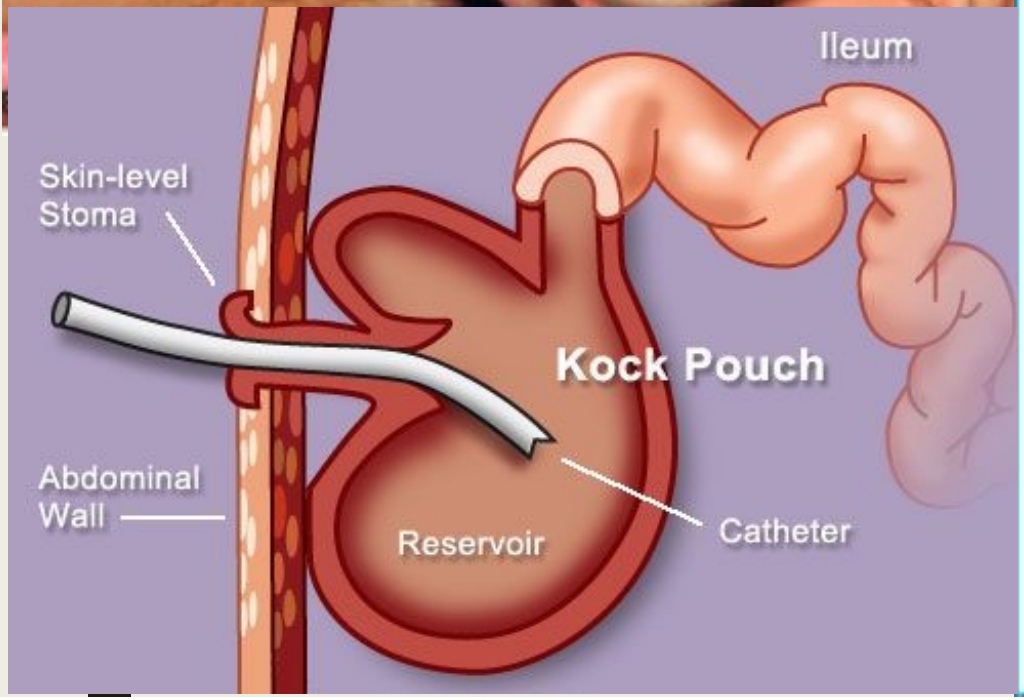
- **Preventing skin breakdown is a major challenge in ostomy care.**
  - ***Enzymes in the stool can quickly cause excoriation (chemical injury of the skin).***
  - ***Washing the stoma and the surrounding skin with mild soap and water, and patting it dry can preserve skin integrity.***
  - ***Another way to protect the skin is to apply barrier substances such as karaya, a plant substance that becomes gelatinous when moistened, and commercial skin preparations around the stoma.***

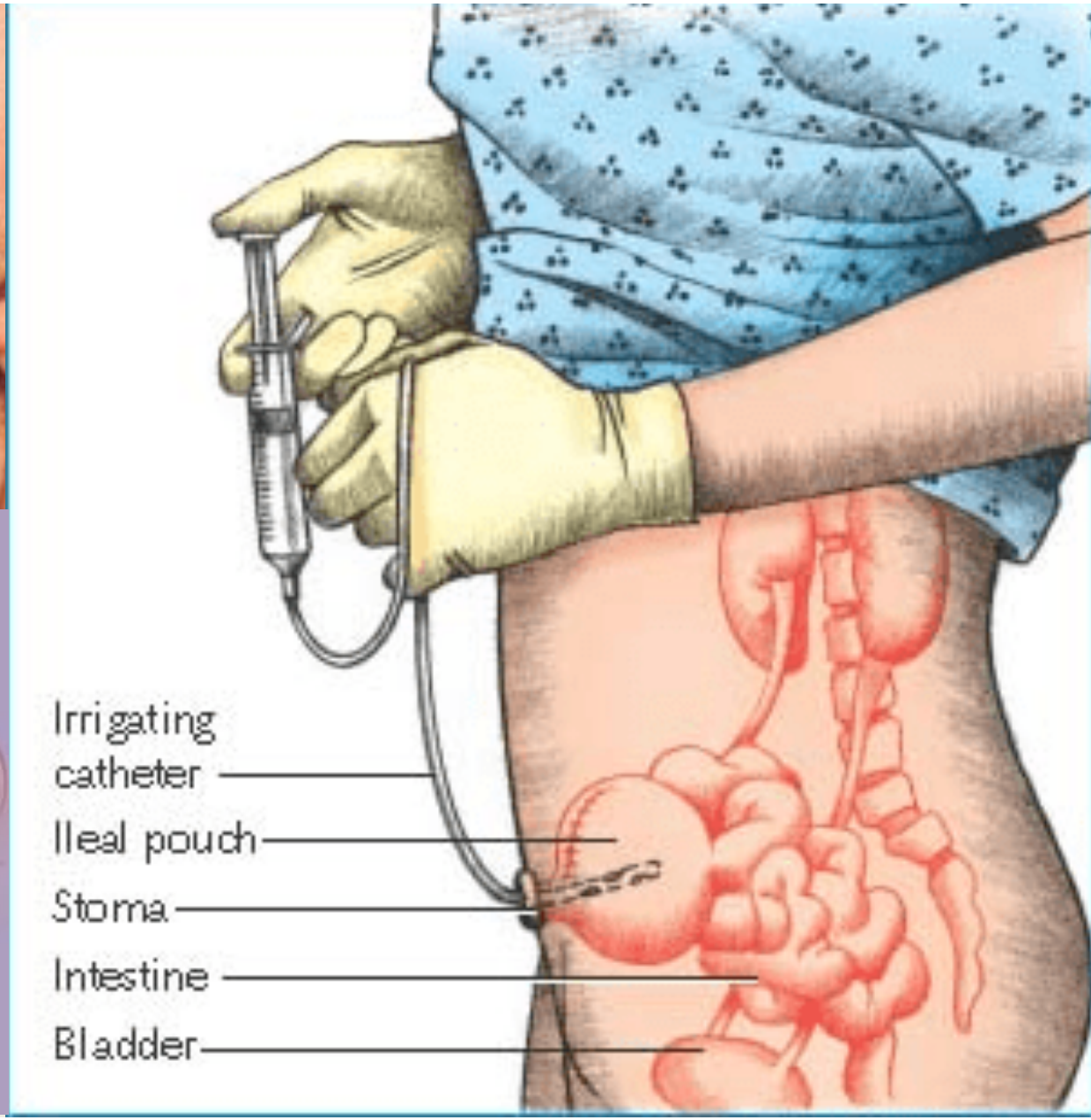
Client and Family Teaching Draining a Continent Ileostomy

- **Draining a Continent Ileostomy**
- A **continent ostomy** (a surgically created opening that controls the drainage of liquid stool or urine by siphoning it from an internal reservoir) also is referred to as a *Kock pouch*, after the surgeon who developed the technique.

- The nurse teaches the client and the family as follows:
  - *Assume a sitting position.*
  - *Insert a lubricated 22- to 28-F catheter into the stoma.*
  - *Expect resistance after inserting the tube approximately 2 inches; this is the location of the valve that controls the retention of liquid stool or urine.*
  - *Gently advance the catheter through the valve at the end of exhalation, while coughing, or while bearing down as if to pass stool.*
  - *Lower the external end of the catheter at least 12 inches below the stoma.*
  - *Direct the end of the catheter into a container or toilet as stool or urine begins to flow.*




Ileum
Skin-level Stoma
Kock Pouch
Abdominal Wall
Reservoir
Catheter


Irrigating catheter
Ileal pouch
Stoma
Intestine
Bladder

Client and Family Teaching Draining a Continent Ileostomy

- **Draining a Continent Ileostomy**
- A **continent ostomy** (a surgically created opening that controls the drainage of liquid stool or urine by siphoning it from an internal reservoir) also is referred to as a *Kock pouch*, after the surgeon who developed the technique.
- The nurse teaches the client and the family as follows:
  - *Allow at least 5 to 10 minutes for complete emptying.*
  - *Remove the catheter and clean it with warm soapy water.*
  - *Place the clean catheter in a sealable plastic bag until its next use.*
  - *Cover the stoma with a gauze square or a large bandage.*
  - *If the catheter becomes plugged with stool or mucus:*
    - Bear down as if to have a bowel movement.
    - Rotate the catheter tip inside the stoma.
    - Milk the catheter.
    - If these are not successful, remove the catheter, rinse it, and try again.
    - Notify the physician if these efforts do not result in drainage.
  - *Never wait longer than 6 hours without obtaining drainage.*


Thank
You
Merci
Obrigado
Asante
Grazie
Maake
Kiitos
Spasibo
Multumesc
Vinaka
Arigato
Matondo
Juspaxar
Dank Je
Nirringrazzjak
Cam on ban
Terma Kasih
Welalin
Mochchakkeram
Ua Tsaug Rau Koj
Kia Ora
Chokrane
Matondo
Dankon
Mamana
Wela
Salamat
Mabur Nuwun
اللہ یحفظک
God Bless You

ORAL MEDICATIONS

Medication

***One of the nurse's most important responsibilities** is the administration of medications.*

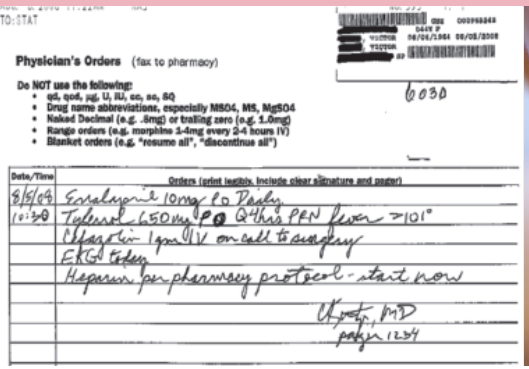
***Medications** (chemical substances that change body function).*

MEDICATION ORDERS

1. A **medication order** lists the drug name and directions for its administration.

- Components of a Medication Order, all medication orders must have seven components:
  1. *The client's name (الاسم رباعي/)*
  2. *The date and time the order is written*
  3. *The drug name (clear name)*
  4. *The dose to be administered (clear and accurate)*
  5. *The route of administration*
  6. *The frequency of administration (eg. 50mg*2*5 or 50mg / 12 hrs.)*
  7. *The signature and the stamp of the person ordering the drug*

NB. Never give medication if the order missing one of its components, or there is any doubt about the information in the order

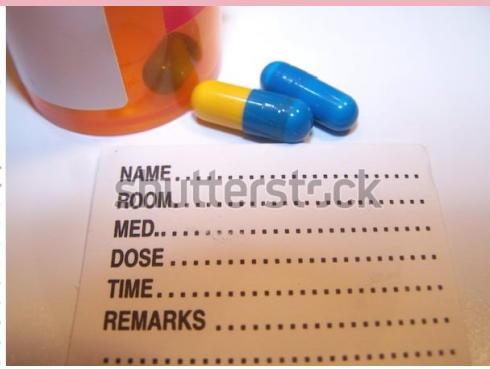

TO: STAT

Physician's Orders (fax to pharmacy)

Do NOT use the following:
- qd, qod, µg, U, IU, cc, sc, SQ
- Drug name abbreviations, especially MSO4, MS, MgSO4
- Naked Decimal (e.g. .5mg) or trailing zero (e.g. 1.0mg)
- Range orders (e.g. morphine 1-4mg every 2-4 hours IV)
- Blanket orders (e.g. "resume all", "discontinue all")

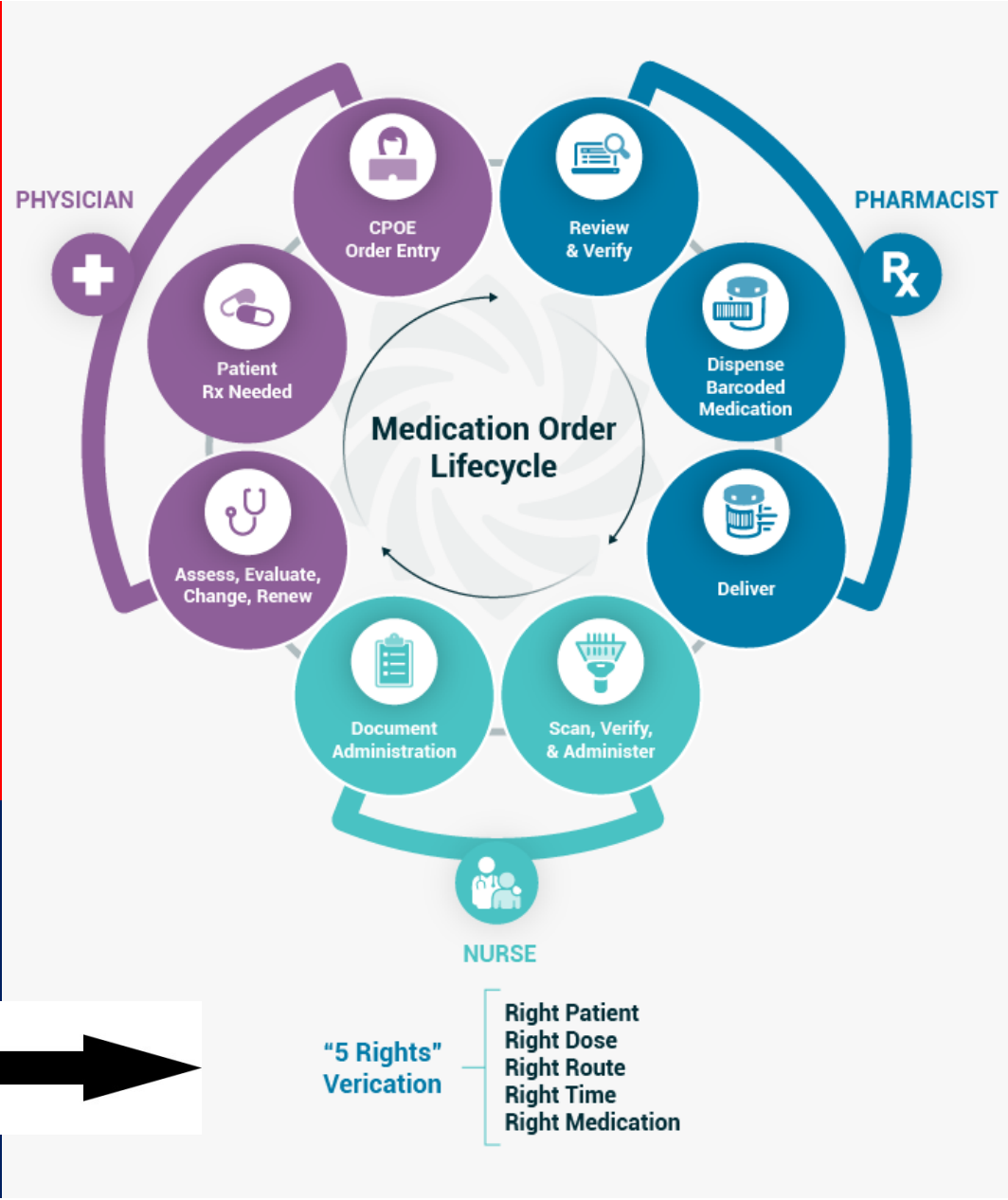
6030

Date/Time	Orders (print legibly. Include clear signature and pager)
8/5/04 10:30	Enalapril 10mg PO Daily
	Tylenol 650mg PO Q4hrs PRN fever >101°
	Ceftriaxone 1gm IV on call to surgery
	EKG today
	Heparin per pharmacy protocol - start now
	Chasty, MD pager 1234



A nurse is legally responsible for ensuring that all components of a medication order are present and correct before administering any drug.
If there is any question in the nurse's mind about the order as it is written, whether it be misleading or difficult to understand, the nurse must get clarification from the person who wrote the order.
Remember, the client's safety is the primary concern.

Patient Five Rights Verification Before Drug administration



Drug Name

Trade / brand name (the name by which a pharmaceutical company identifies its drug).
A drug's trade name is generally capitalized and followed by an R within a circle, as in ®.
Generic name (a chemical name not protected by a company's trademark), which is written in lowercase letters. which is written in lowercase letters.

For example, Demerol® is a trade name used by Winthrop Pharmaceuticals for the generically named drug meperidine hydrochloride.

Drug Dose

- *The dose means the amount of drug to administer and is prescribed using the metric system.*
- ***Nurse responsibility to calculate the quantity / volume accurately to have the desired (ordered) dose including in the calculated volume***

Dose in hand (H)		*Desired dose (D)*
Volume from supplier (V)		*Needed Volume /quantity (??)*

Basic dosage calculation

$$\frac{\text{D (desired dose)}}{\text{H (amount on hand)}} \times \text{V (volume)} = \text{Dose}$$

$$\frac{\text{Desired amount or prescribed amount (D)}}{\text{Available dosage (what's on hand)(A)}} \times \text{Quantity (e.g. 1 pill or 1 tablet or 5 mL} = X)$$

$$\frac{\text{Desired amount}}{\text{Available dosage}} \times \text{Quantity} = \text{amount to give}$$

Drug Calculation Formula

$$\frac{D}{H} \times Q = \frac{\text{Desired dose}}{\text{Dose on hand (supplied dose)}} \times \text{Quantity}$$
$$= \text{Amount to administer}$$

Example

Drug order: Tetracycline 500 mg (desired dose) by mouth q.i.d.

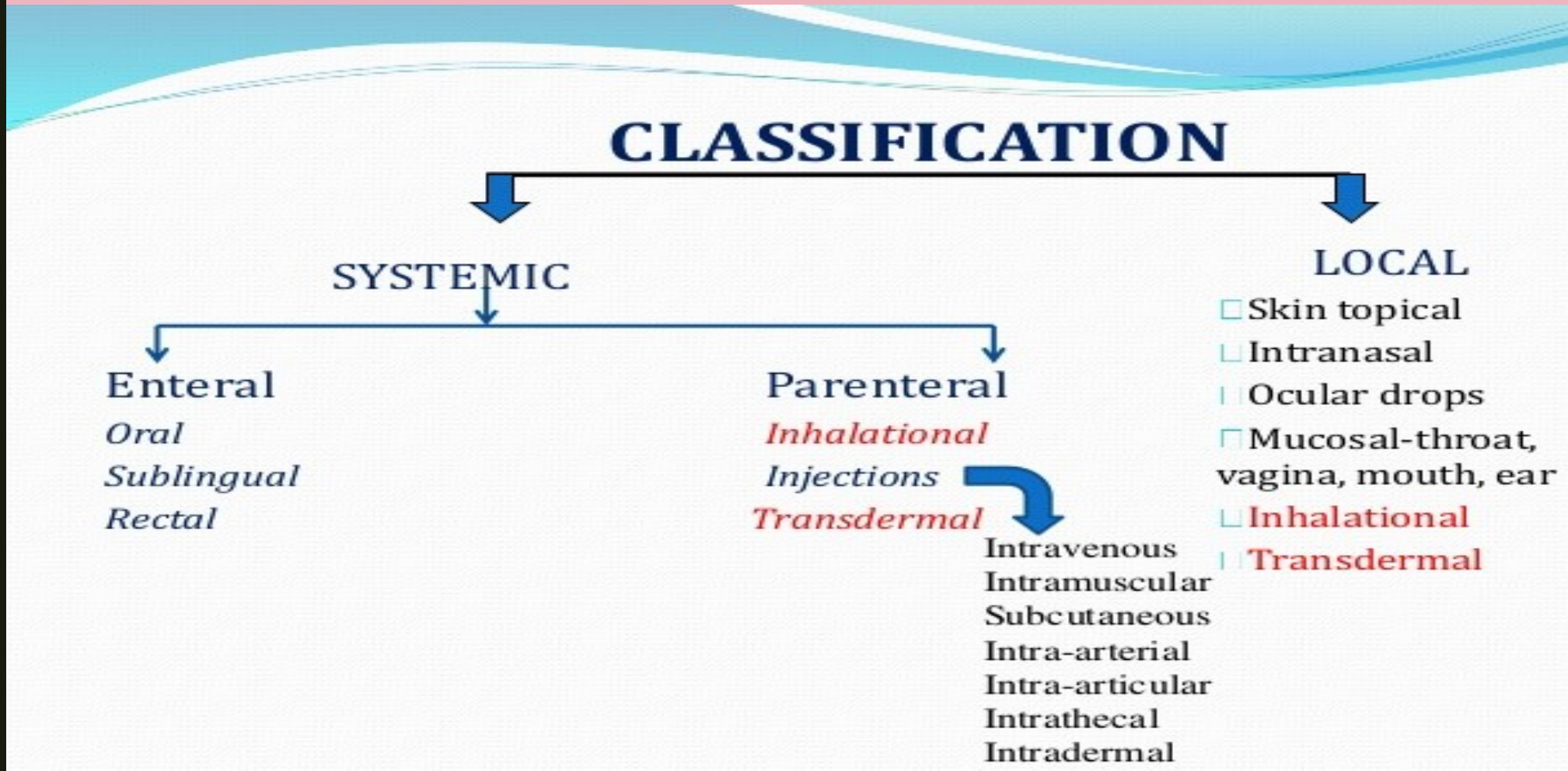
Dose supplied: 250 mg (dose on hand) per 5 mL (quantity)

$$\text{Calculation: } \frac{500 \text{ mg}}{250 \text{ mg}} \times 5 \text{ mL} = 10 \text{ mL}$$

Routes of Administration

- **Factors Deciding Choice of Route**
  - Type of described effect, systemic or local
  - Physiochemical properties, solid or insoluble
  - Rapidity of effect
  - Quality of effect
  - Condition of patient, conscious or unconscious, vomiting

Routes of Administration



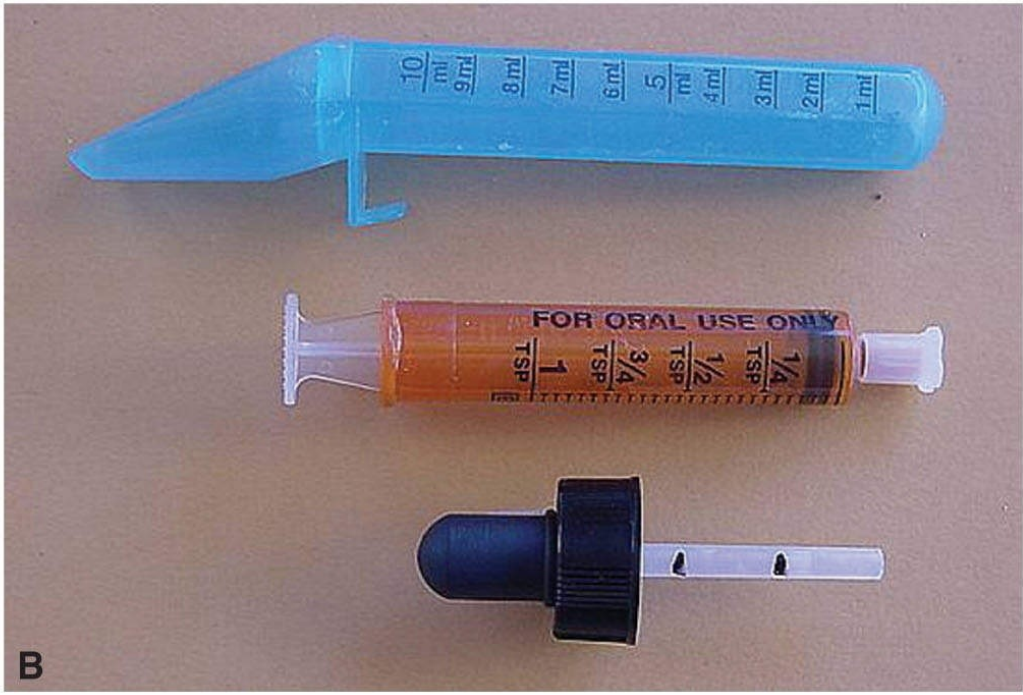
Oral Rout

- **The oral route** (the administration of drugs by swallowing or instillation through an enteral tube).
  - It is the most common route for medication administration
  - it is safer, more economical, and more comfortable than others.
  - Oral Medications come in both solid and liquid forms.
  - Solid medications include:
    - tablets and
    - capsules.
  - Enteric-coated tablets (a solid drug covered with a substance
    - that dissolves beyond the stomach)
  - Some tablets and capsules may be identified as
    - sustained release (SR), which indicates that the drug is designed to dissolve slowly and be released over time.

Oral Rout

- **Liquid forms of oral drugs include :**
  - syrups, elixirs, and suspensions.
  - Nurses measure and administer liquid medications in calibrated cups, droppers, or syringes, or with a dosing spoon.





Frequency of Administration

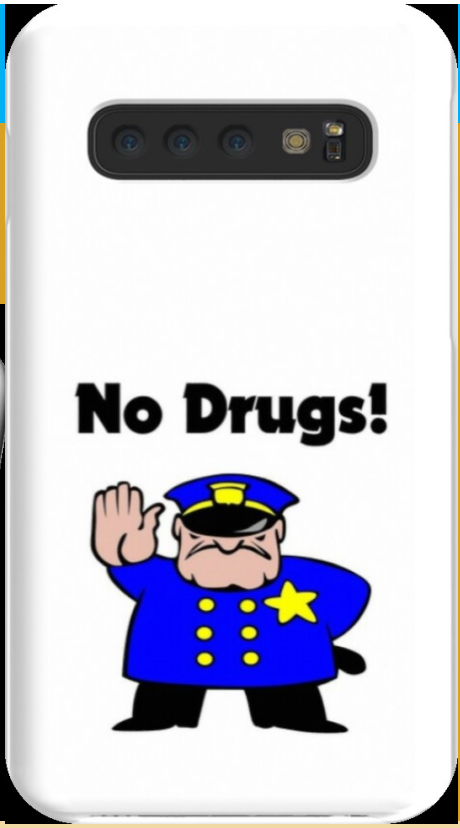
Frequency of administration is written using standard abbreviations of Latin origin. Some common examples include the following:

Latin abb.	Or like this	Or like this
Stat—immediately	Immediately once	Immediately once
b.i.d.—twice a day	Dose / 12 hours	1*2
t.i.d.—three times a day	Dose / 8 hours	1*3
q.i.d.—four times a day	Dose / 6 hours	1*4
q.h.—hourly	Dose / 1 hours	1*12
q4h—every 4 hours	Dose / 4 hours	1*6
p.r.n.—as needed	SOS	SOS

Verbal and Telephone Orders

Never, never, and never administer medications by verbal orders in ordinary circumstances





- **Circumstances when verbal and telephone orders are justified:**
  1. *When a physician is involved in a procedure he or she cannot leave.*
  2. *When a delay in obtaining an order would cause harm to a client.*
  3. *When an order is urgently needed in less time than it would take to physician arrival.*

Verbal and Telephone Orders

- **Circumstances when verbal and telephone orders are justified:**
  1. *Always identify yourself, your agency, and unit on which you are working.*
  2. *Identify the client, the situation, background circumstances, pertinent assessments, and recommendation (SBAR) concerning care.*
  3. *Make sure the order includes the essential components.*
  4. *Clarify or spell drug names for those that sound similar, such as Celebrex and Cerebrex.*
  5. *Spell numbers to avoid misinterpretation, such as 15 (one, five) and 50 (five, zero).*
  6. *Read back the written information to the prescriber.*
  7. *Use the abbreviation "T.O." at the end of the order.*
  8. *Write the prescriber's name and cosign with your name and title.*
  9. *Remind the prescriber that the order must be signed as soon as possible.*

Preparing Medications Safely

- **Circumstances when verbal and telephone orders are justified:**
  1. *Prepare medications under well-lit conditions.*
  2. *Work alone without interruptions and distractions.*
  3. *Check the label of the drug container three times:*
     - ✓ (1) when reaching for the medication,
     - ✓ (2) just before placing the medication into an administration cup, and
     - ✓ (3) when returning the medication to the
  4. *Avoid using medications from containers with a missing or obliterated label.*
  5. *Return medications with dubious or obscured labels to the pharmacy.*
  6. *Never transfer medications from one container to another.*
  7. *Check the expiration dates on liquid medications.*
  8. *Inspect the medication and reject any that appear to be decomposing.*

General procedures for oral administration

- Identify the medication order, use hospital policy when checking medication orders
- Wash your hands and prepare the medication
- Identify your client and explain the medications to be given
- Take the required assessment measures (B/P, apical pulse, etc)
- Assist the client to take the medications
- Stay with the client until all medications have been taken
- Document the medication administration
- Evaluate the effects of the medication
- Report any undesired responses or side effects.

Preparing Medications for Enteral Tube Administration

- Use the liquid form of the drug whenever possible.
- Add 15 to 60 mL of water to thick liquid medications.
- Crushing solid oral medications. Open the shell of a capsule to release the powdered drug.
- Avoid crushing sustained-release pellets.
- Mix each drug separately with at least 15 to 30 mL of warm water.
- Interrupt a tube feeding for 15 to 30 minutes before and after administration of a drug
- Flush tube by 15-30 ml of water after drug administration.

When the client is planning to go home

- Be specific with instructions on how to take and work out a schedule that fits around the client's home routine
- Instruct the client on what to do if a dose is missed
- Emphasize the importance of not discontinuing the medication without consulting his/her doctor
- Emphasize the importance of not sharing medications with others who may have similar symptoms
- Write the instructions out for the client – in clear language
- Have the client repeat the instructions back to you

Medication Errors

1. As soon as an error is recognized:
   - ✓ the nurse checks the client's Work alone without interruptions and distractions.
   - ✓ reports the mistake to the prescriber and the supervising nurse immediately.
   - ✓ Health care agencies have a form for reporting medication errors called an incident sheet, or accident sheet.


Thank
You
Merci
Obrigado
Asante
Grazie
Maake
Kiitos
Spasibo
Multumesc
Vinaka
Arigato
Matondo
Juspaxar
Dank Je
Nirringrazzjak
Cam on ban
Terma Kasih
Welalin
Mochchakkeram
Ua Tsaug Rau Koj
Kia Ora
Chokrane
Matondo
Dankon
Mamana
Wela
Salamat
Mabur Nuwun
اللہ یحفظک
God Bless You

TOPICAL AND INHALANT MEDICATIONS

Topical and Inhalant Medications

Drugs are administered by routes other than oral

TOPICAL ROUTE

- Drugs given by the **topical route** (the administration of medications to the skin or mucous membranes) can be applied externally or internally.
- Topically applied drugs have a local or systemic effect.

Percutaneous Applications

- Percutaneous applications are drugs rubbed into or placed in contact with the skin. They include:
- ointments,
  - *An ointment is a topical preparation rubbed into the skin for administration of a medication (also called oils, lotions, and creams).*
- patches, and
  - *Skin patches are drugs bonded to an adhesive and applied to the skin for systemic distribution.*
- pastes.
  - *A paste contains a drug within a thick base and is applied to but not rubbed into the skin.*

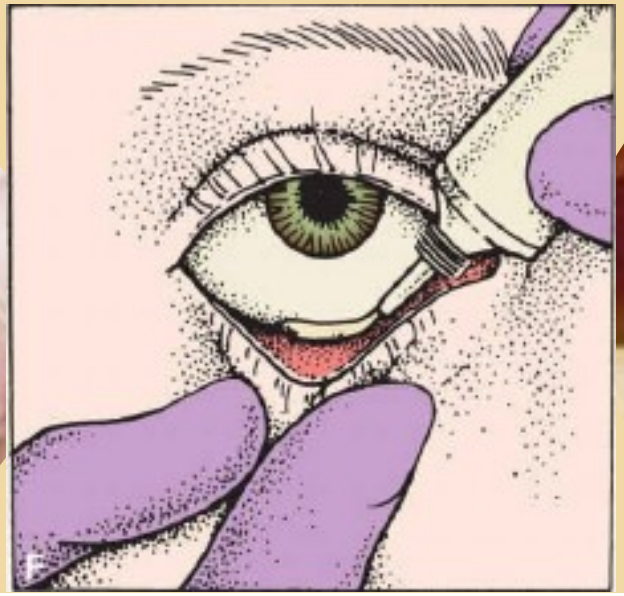
Applying a Percutaneous Application

- Wash your hands or perform an alcohol-based hand rub.
- Check the identity of the client.
- Put on clean gloves if your skin or that of the client is not intact.
- Cleanse the area of application with soap and water.
- Warm the substance if it will be applied to a sensitive area.
- Shake the contents of liquids.
- Apply the percutaneous substance to the skin with the fingertips, a cotton ball, or a gauze square.
- Rub the substance into the skin. Rubbing promotes absorption.
- Apply local heat to the area if desired. Heat dilates peripheral blood vessels and speeds absorption.

Ophthalmic Applications

- An ophthalmic application is a method of applying drugs onto the mucous membrane of one or both eyes. The mucous membrane of the eyes is called the conjunctiva.
  - Ophthalmic medications are supplied either in liquid form and instilled as drops, or as ointments applied along the lower lid margin.
  - Blinking, rather than rubbing, distributes the drug over the surface of the eye.





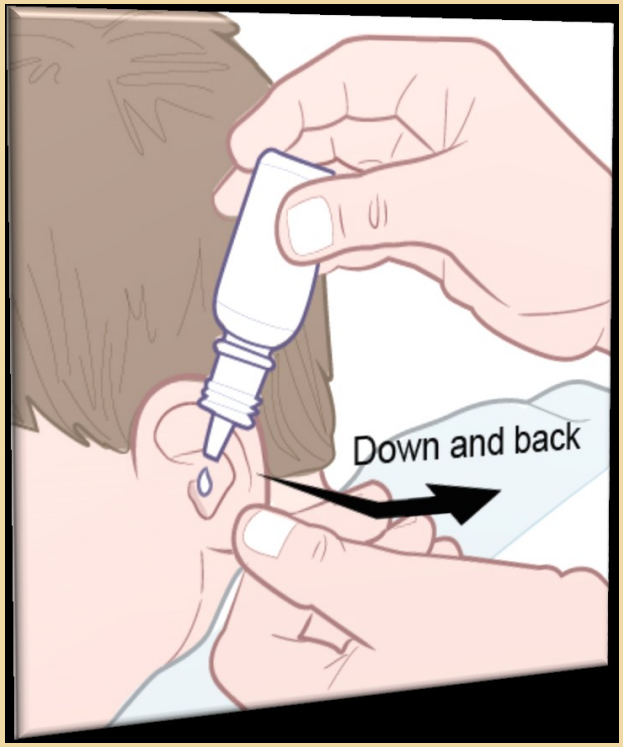


Ophthalmic Applications

- An ophthalmic application is a method of applying drugs onto the mucous membrane of one or both eyes. The mucous membrane of the eyes is called the conjunctiva.
  - Wash your hands with soap and water.
  - Remove the dropper cap.
  - Either lie client down or tilt head back, and looking up at the ceiling.
  - Place one or two fingers on client face an inch below eye, pull down to make a pouch in lower lid.
  - Use your other hand to hold the eye drop bottle, pointing the tip downward.
  - Hold the bottle close to eye, squeeze lightly to instill one drop inside lower lid.
  - Use a clean tissue to absorb and wipe from in to out side “from inner canthus”
  - Replace the cap of the bottle and screw it on securely.
  - Wash your hands to rinse away any stray eye drops.
  - Waite 5 minutes before applying other different medication.

Otic Applications

- Position the head so that the ear faces upward.
- For adults, gently pull the ear up & back. For children, gently pull the lower ear down and back.
- Gently pull the earlobe up and down to allow the drops to run into ear.
- Wipe away any extra liquid with a tissue or clean cloth.



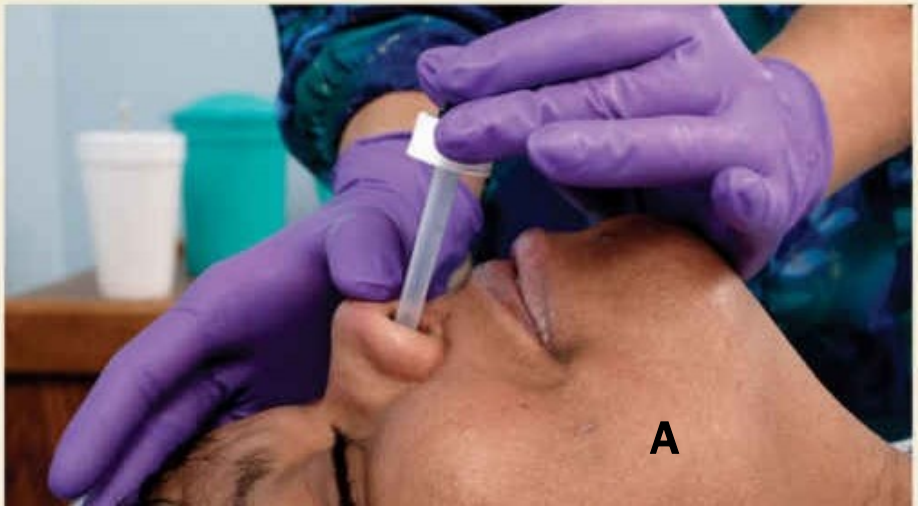
- Insert a cotton ball loosely with the opening to the ear canal.
- Put the cap back on the bottle.
- Wash your hands to remove any medication.
- Store the bottle as directed by the label or pharmacist.





Nasal Applications

Wash your hands or perform an alcohol-based hand rub (see Chap. 10).
Identify the client using at least two methods, such as scanning or reading the wristband, and asking the client's name and birthdate.
Help the client to a sitting or lying position with his or her head tilted backward or to the side if the drug needs to reach one or the other sinuses.
Place a rolled towel or pillow beneath the neck if the client cannot sit.
Remove the cap from the liquid medication exposing the outlet for the medication.
Aim the tip of the container or dropper toward the nasal passage and squeeze to administer the number of drops prescribed (Fig. A).


A

Instruct the client to breathe through the mouth as the drops are instilled.
If the drug is in a spray form, place the tip of the container just inside the nostril (Fig. B).


B

Occlude the opposite nostril.
Instruct the client to inhale as the container is squeezed.
Repeat in the opposite nostril.

Ophthalmic Applications

- **Sublingual**
  - *Medication is placed under the tongue until absorbed*
  - *Drug is mainly absorbed into the blood vessels under tongue*
  - *Caution client not to swallow*

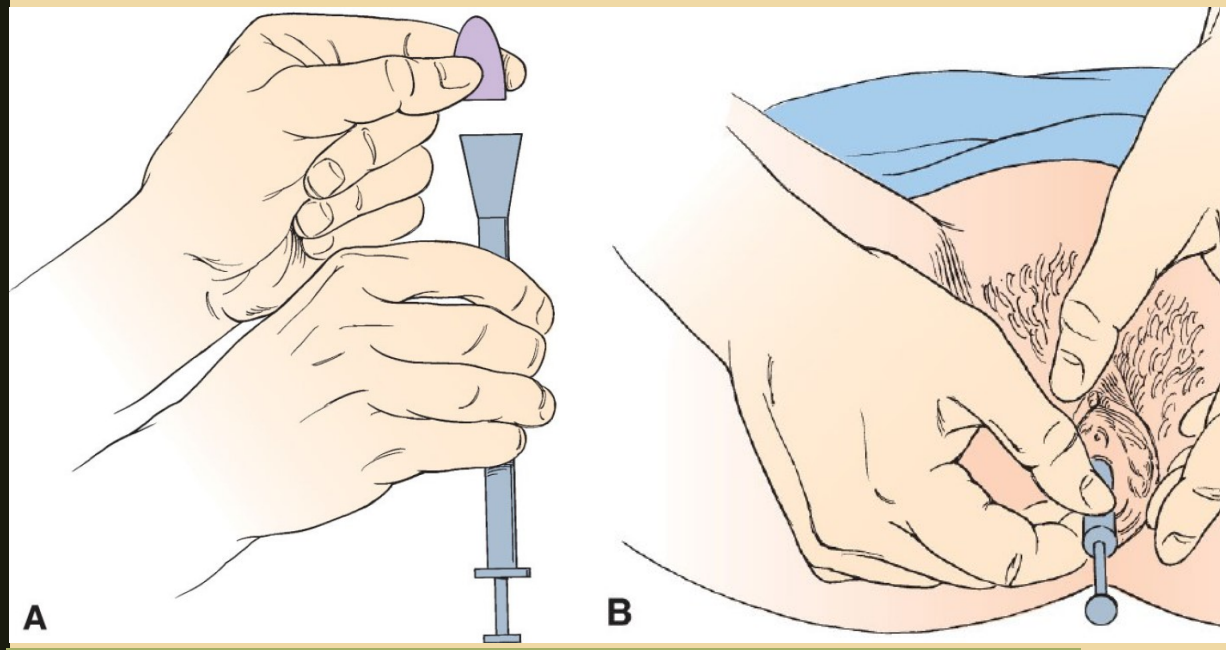


- ***Buccal***
  - *Medication is placed between the cheek and the teeth and held there until absorbed*
  - *Medication is in tablet form*
  - *Caution client not to swallow*

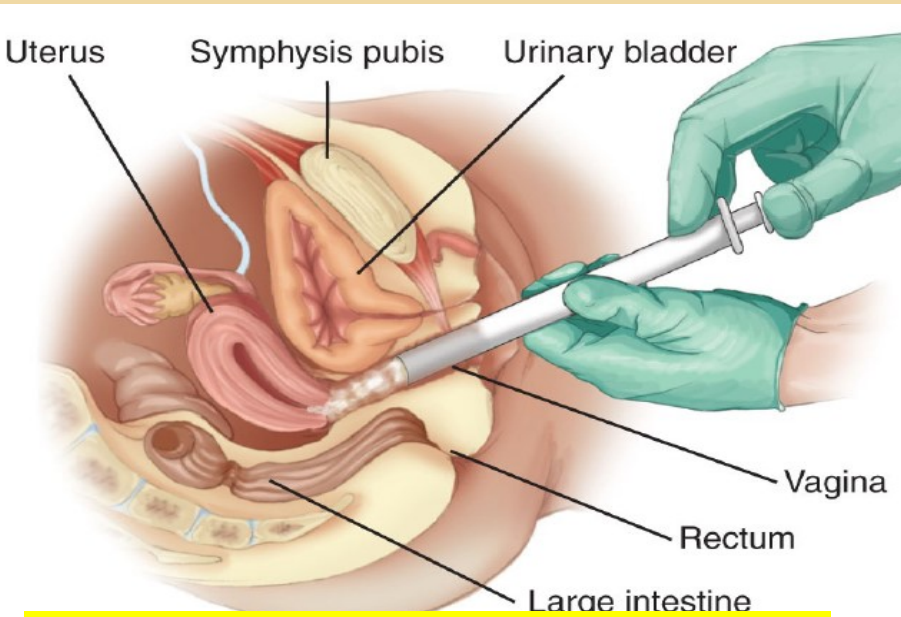


Vaginal Applications

- **Vaginal Applications**
  - *Topical applications that are used most often to treat local vaginal infections.*



A- Inserting the applicator with the drug within the vagina
B- Placing the vaginal drug in the applicator



Insertion of vaginal medication by a nurse.

Rectal Applications

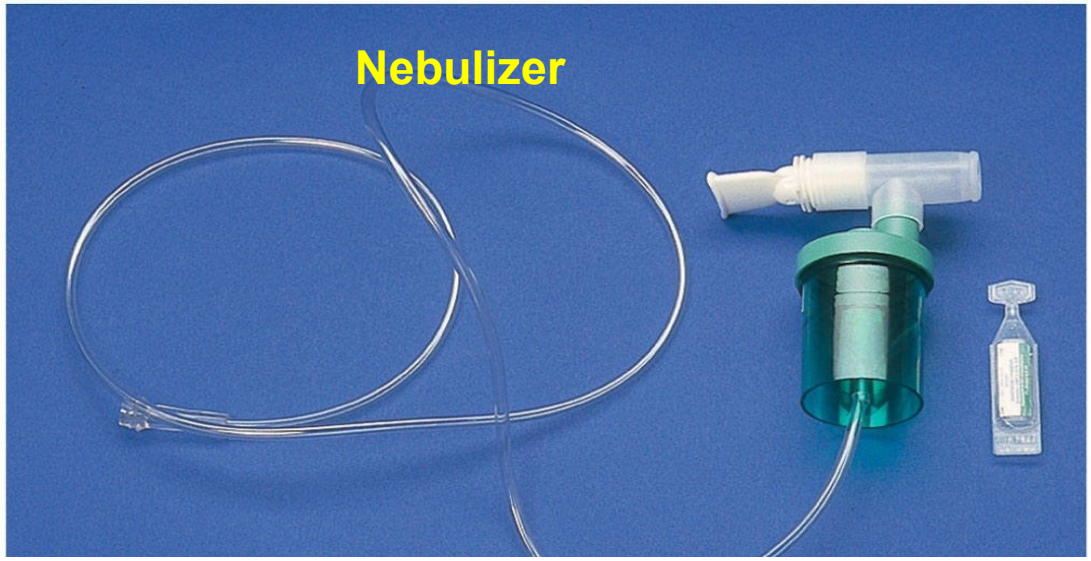
- *Drugs administered rectally are usually in the form of suppositories*
- *Creams and ointments also may be prescribed.*

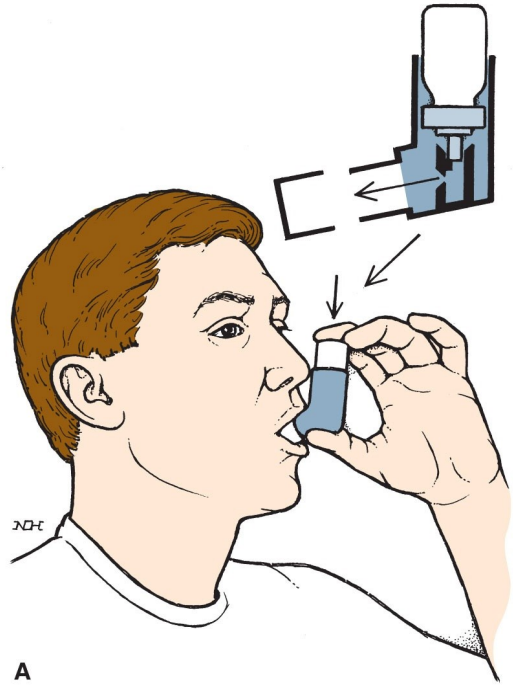
1. *Wash your hands thoroughly with soap and worm water.*
2. *Done clean gloves.*
3. *Client should be lying on their side or front.*
4. *Hold one buttock gently to one side so that you can see the rectum.*
5. *Unwrap the suppository and hold it with the rounded end close to the rectum.*
6. *Use one finger to push the suppository gently into the rectum. It needs to go in by about 2 cm.*
7. Client should stay lying on their side for about 15 ms. This will allow the ttt to spread further inside the bowel and make sure it doesn't come out.
8. Wash your hands again with soap and worm water.

INHALANT ROUTE

- The **inhalant route** administers drugs to the lower airways. This method of medication administration is effective because the lungs provide an extensive area from which the circulatory system can quickly absorb the drug.
- There are two types of inhalers:
  - *(1) a **dry powder inhaler** holds a reservoir of pulverized drug and a carrier substance, and*
  - *(2) a **metered-dose inhaler** that delivers aerosolized medication, which is a liquid drug forced through a narrow Channel via a chemical propellant.*
- A nebulizer, sometimes called a “breathing machine,” is a device that converts liquid medication to an aerosol using compressed air.









A. As the medication is released, breathe in slowly through your mouth for approximately 3 to 5 seconds.
B. Hold your breath for 10 seconds to let the medication reach your lungs.

Medication Errors

1. As soon as an error is recognized:
   - ✓ the nurse checks the client's Work alone without interruptions and distractions.
   - ✓ reports the mistake to the prescriber and the supervising nurse immediately.
   - ✓ Health care agencies have a form for reporting medication errors called an incident sheet, or accident sheet.


Thank
You
Merci
Obrigado
Asante
Grazie
Maake
Kiitos
Spasibo
Multumesc
Vinaka
Arigato
Matondo
Juspaxar
Dank Je
Nirringrazzjak
Cam on ban
Terma Kasih
Welalin
Mochchakkeram
Ua Tsaug Rau Koj
Kia Ora
Chokrane
Matondo
Dankon
Mamana
Wela
Salamat
Mabur Nuwun
اللہ یحفظک
God Bless You

PARENTERAL MEDICATIONS

Parenteral route

Parenteral route is a route of drug administration other than oral or through the gastrointestinal tract.

"Parenteral" is commonly used when referring to medications given by injection.

- When the parenteral route is used for drug administration the drug filtering performed by the liver does not occur.
- Clients, particularly children and older adults, should be monitored closely for drug effects.